

BFS Traversal

Step 1: Start at A

Legend

Complete

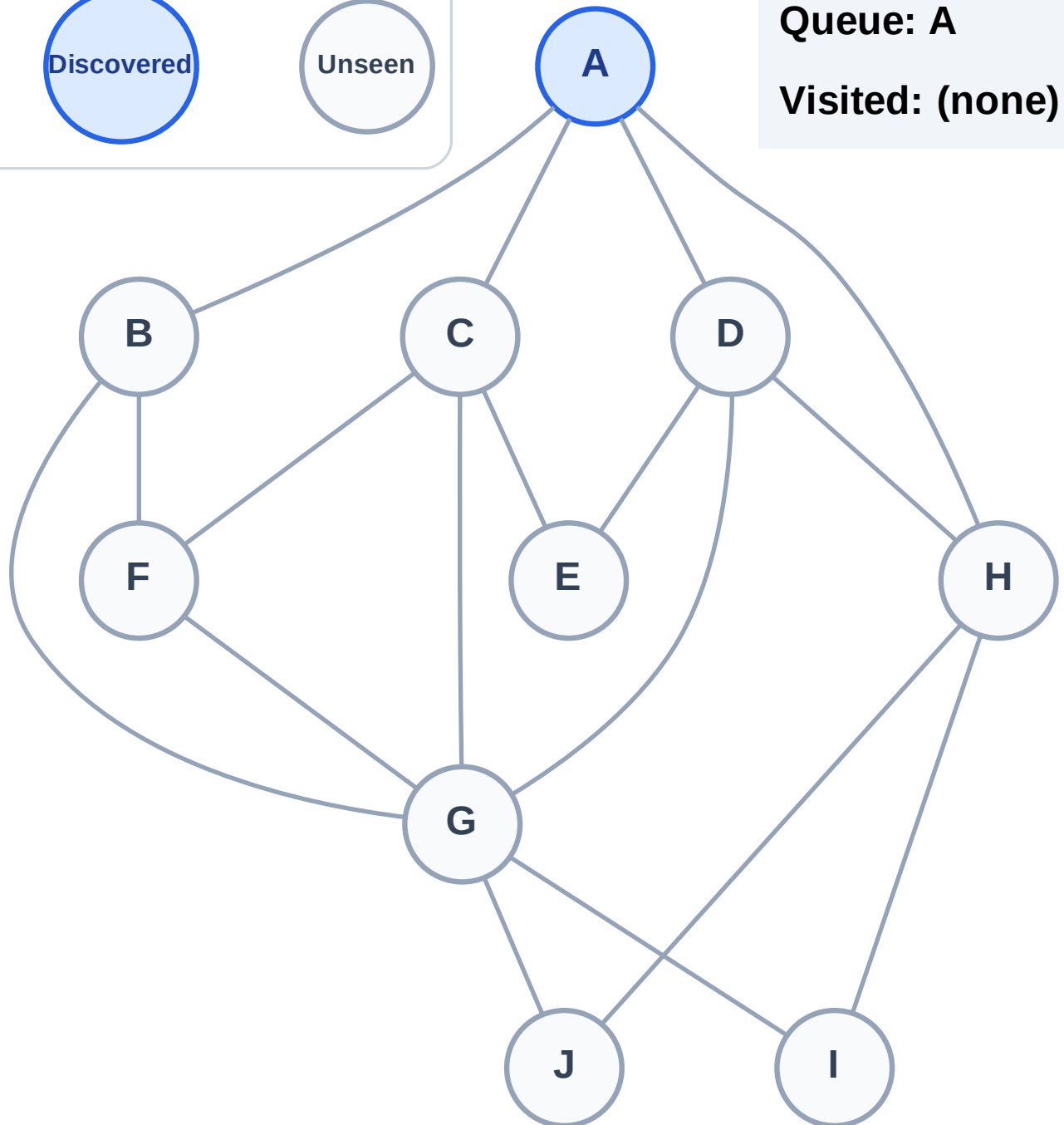
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

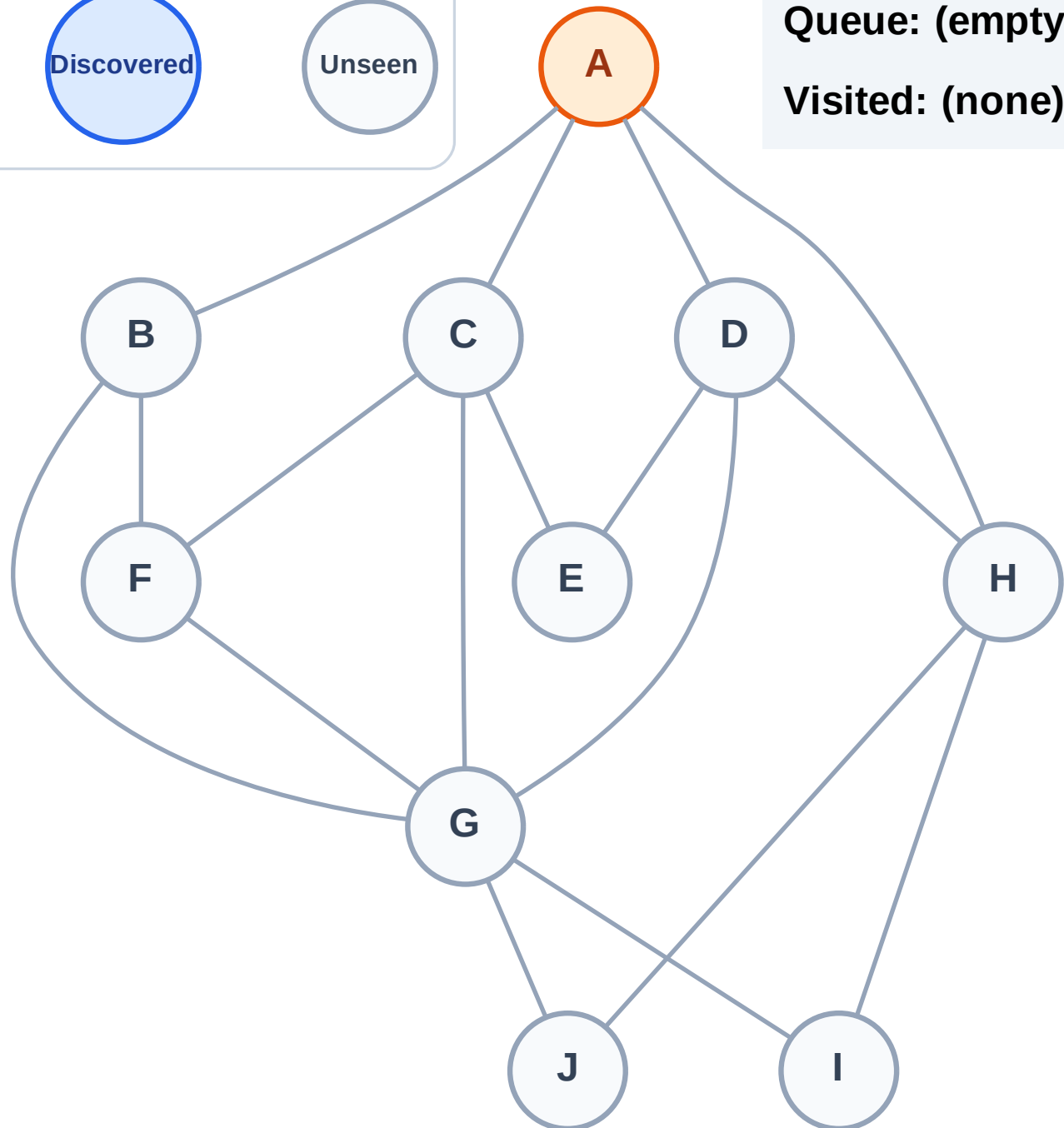
Step 2: Process node A

Legend



Queue: (empty)

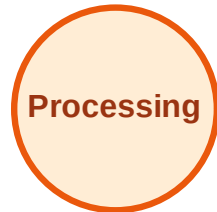
Visited: (none)



BFS Traversal

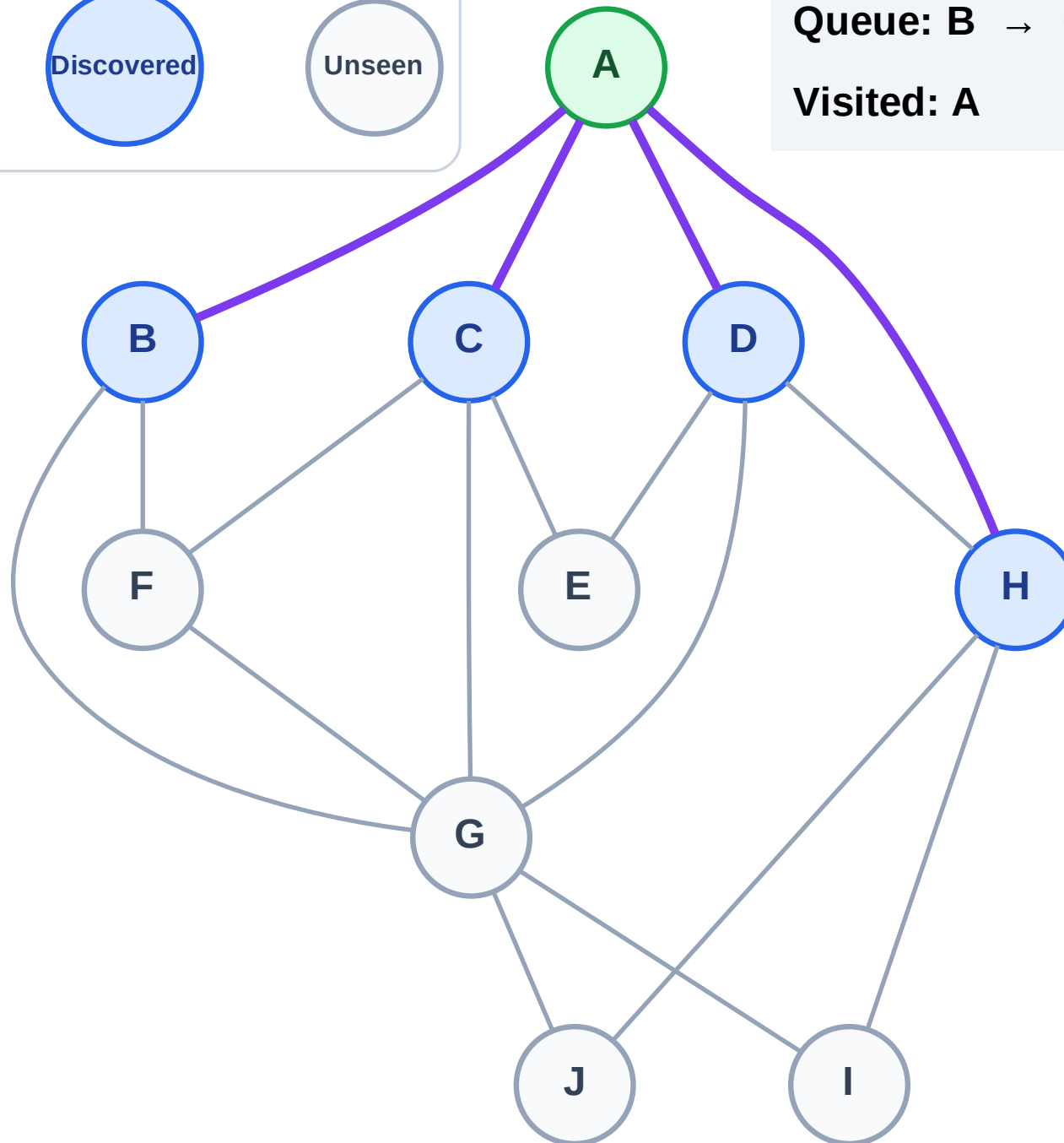
Step 3: Discover B, C, D, H from A

Legend



Queue: B → C → D → H

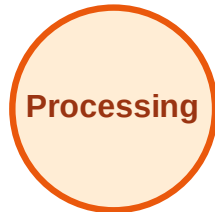
Visited: A



BFS Traversal

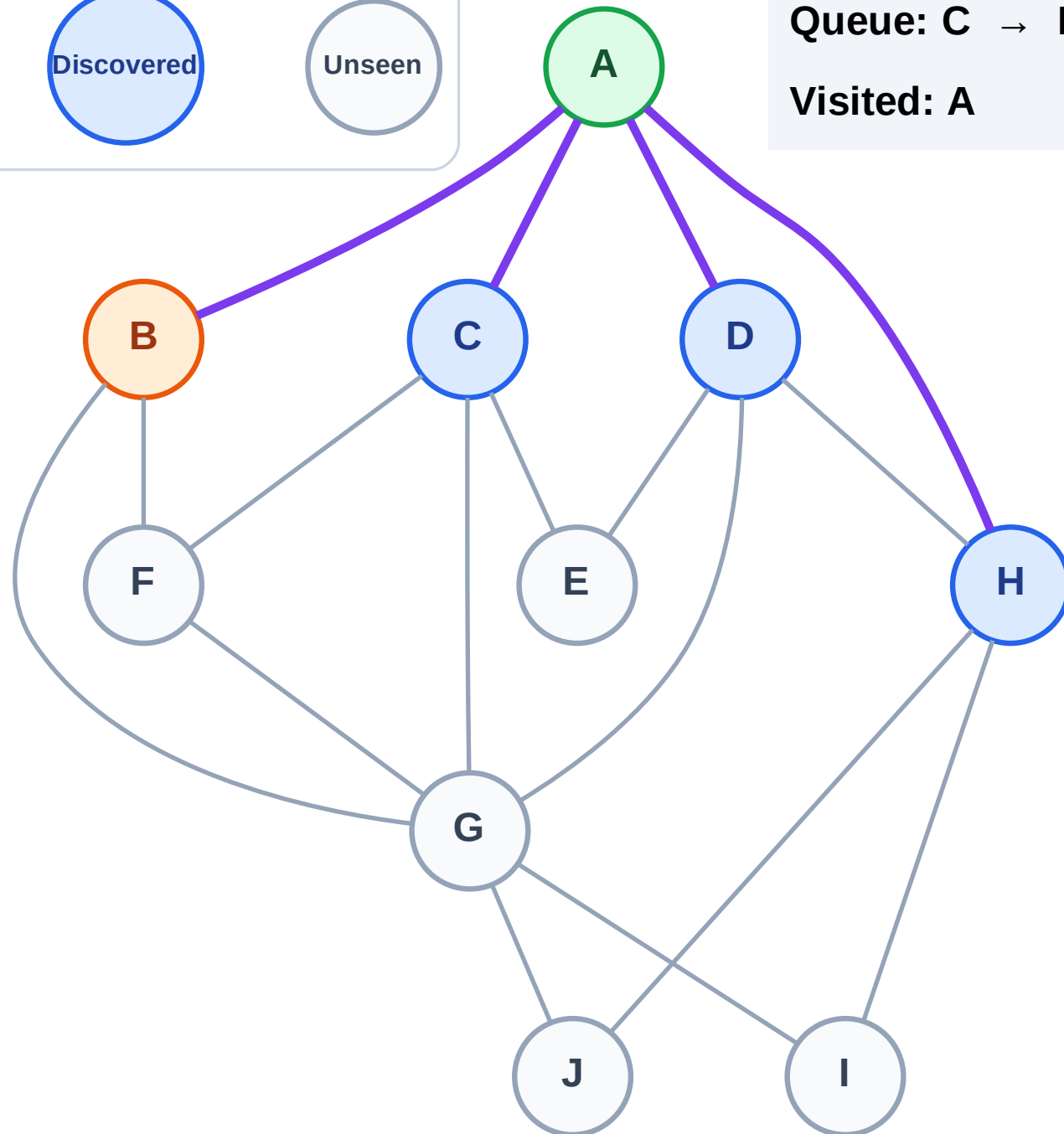
Step 4: Process node B

Legend



Queue: C → D → H

Visited: A



BFS Traversal

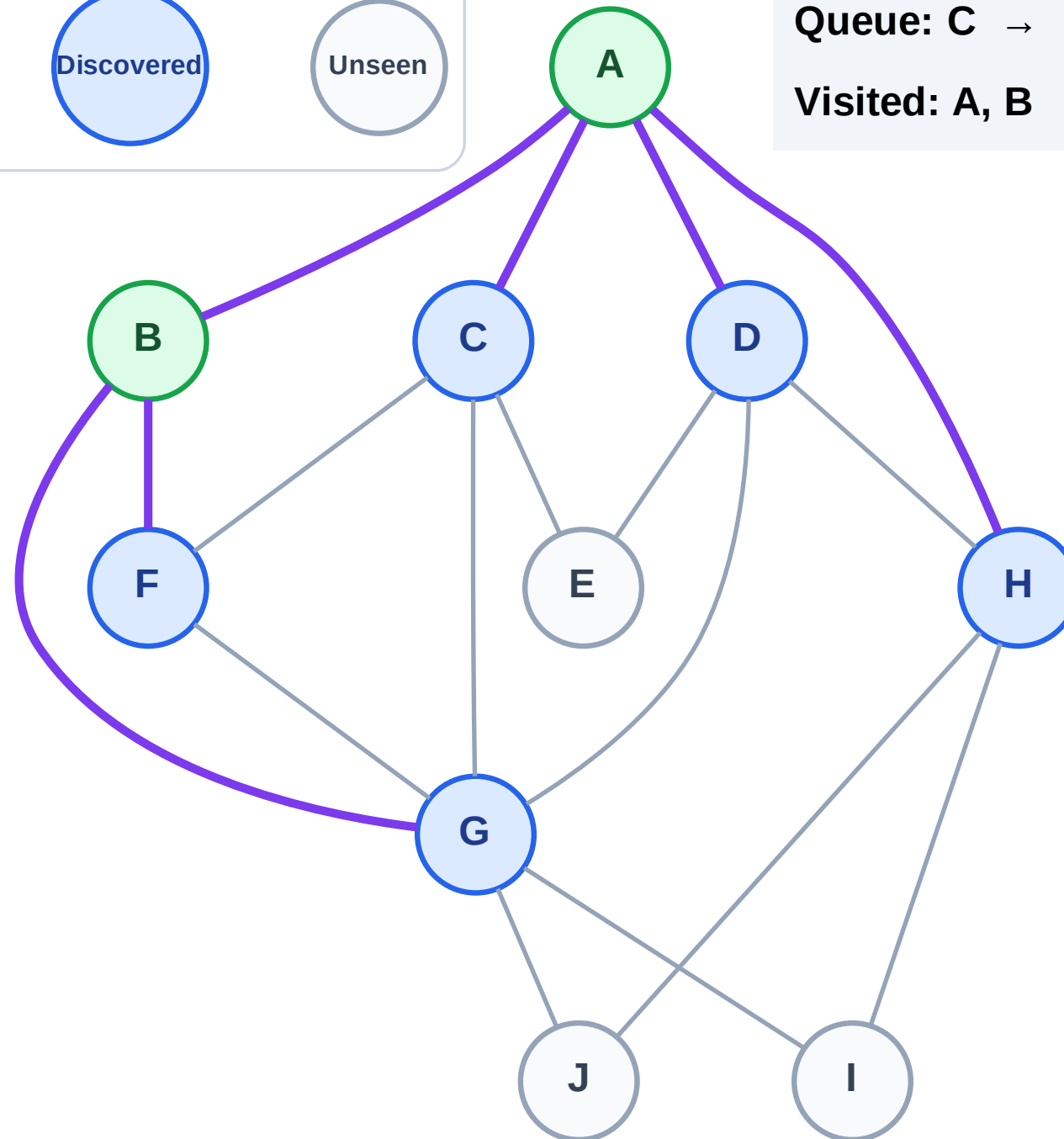
Step 5: Discover F, G from B

Legend



Queue: C → D → H → F → G

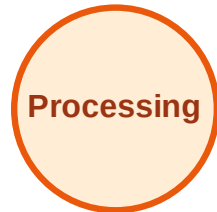
Visited: A, B



BFS Traversal

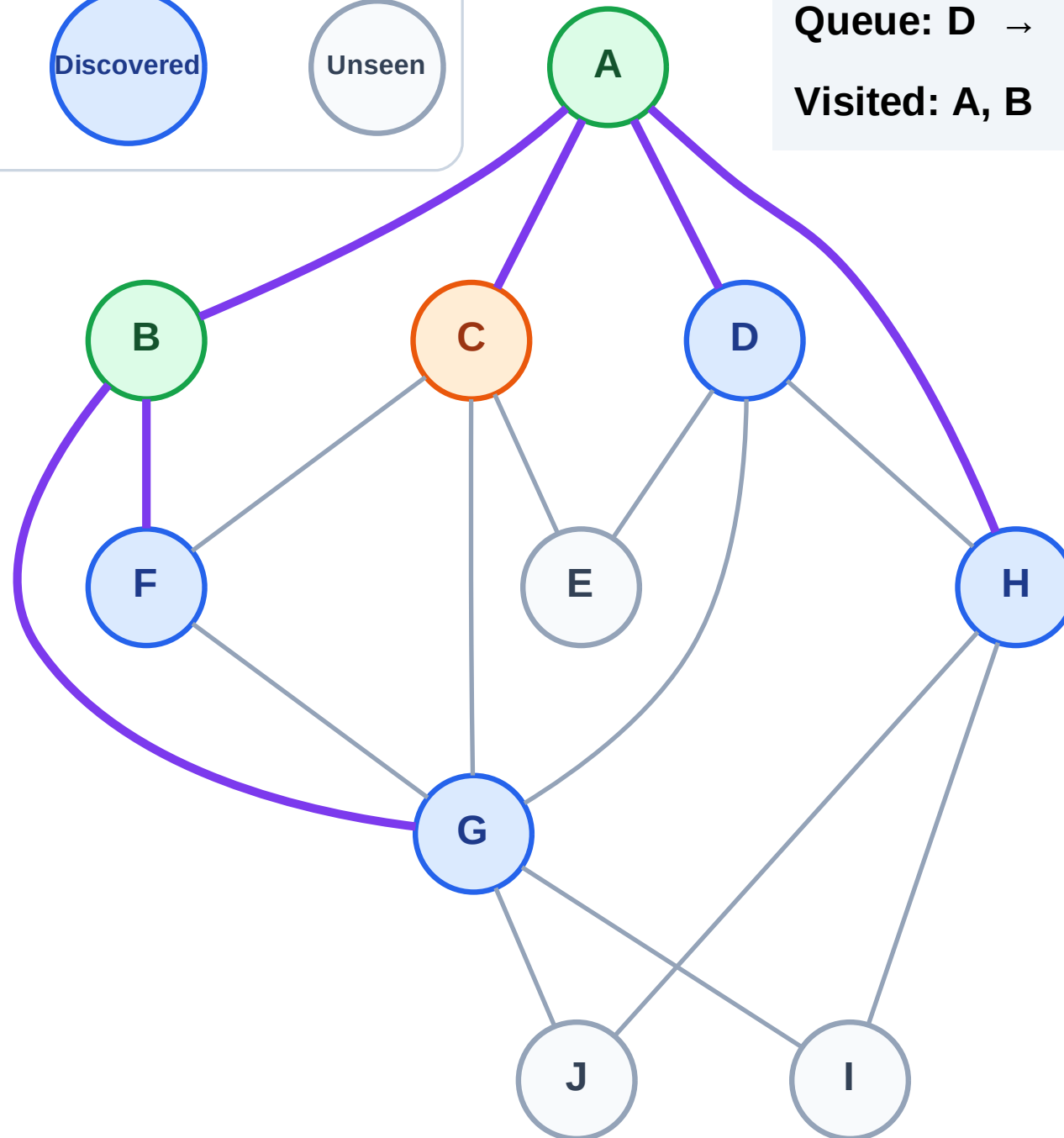
Step 6: Process node C

Legend



Queue: D → H → F → G

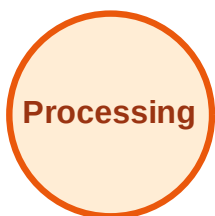
Visited: A, B



BFS Traversal

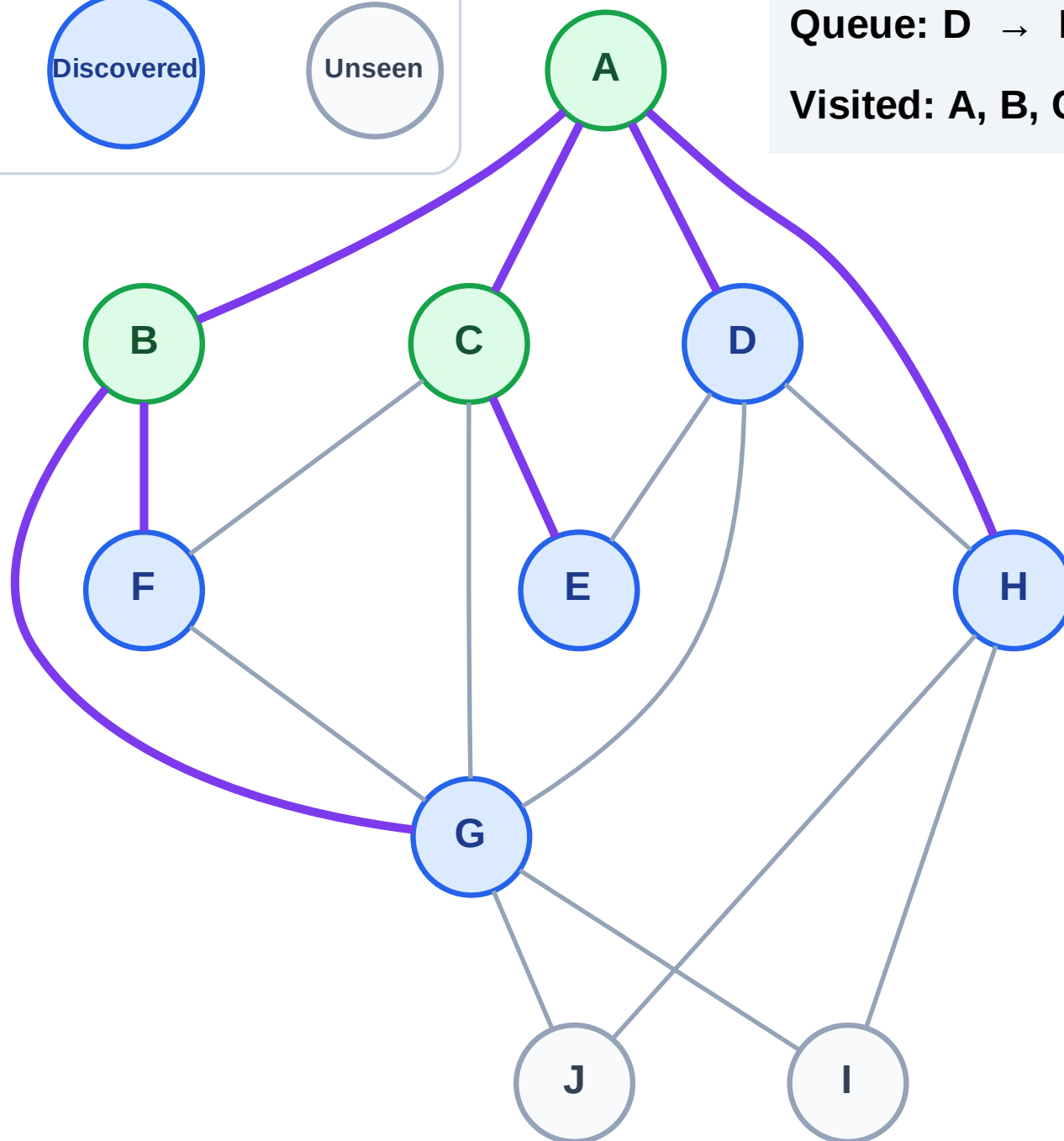
Step 7: Discover E from C

Legend



Queue: D → H → F → G → E

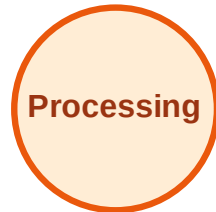
Visited: A, B, C



BFS Traversal

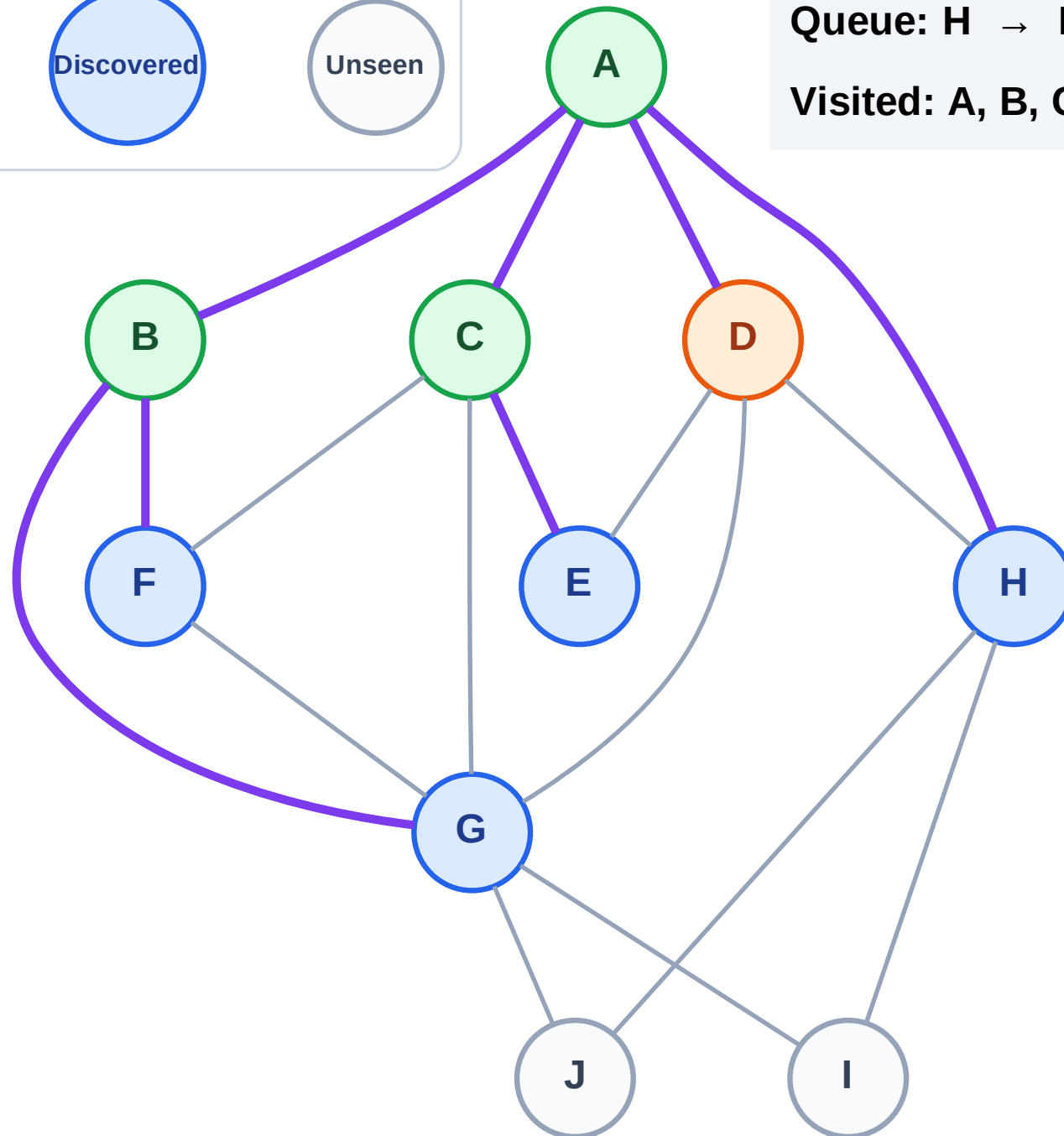
Step 8: Process node D

Legend



Queue: H → F → G → E

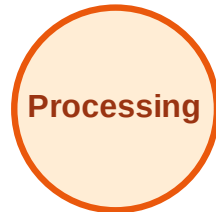
Visited: A, B, C



BFS Traversal

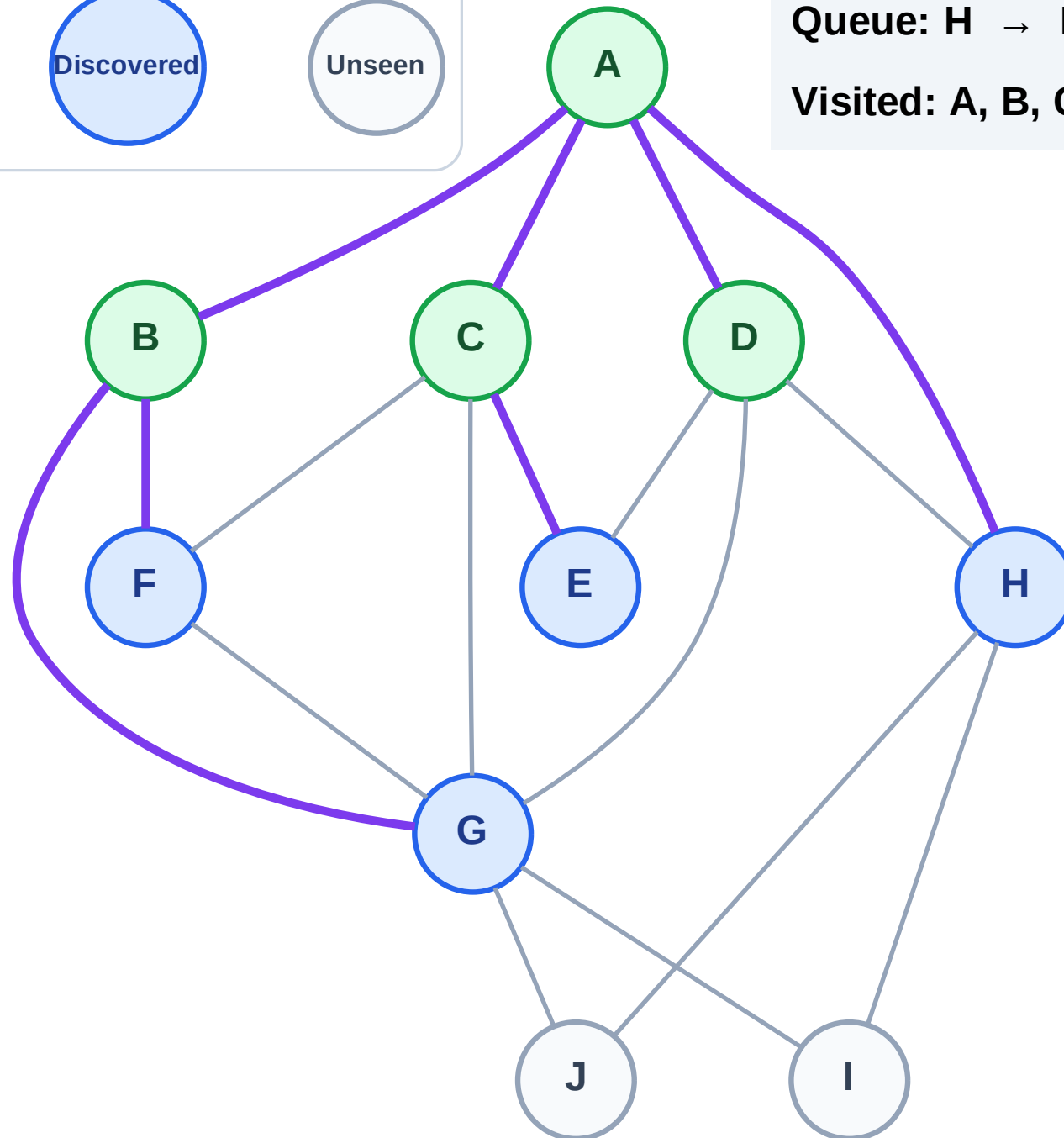
Step 9: No new neighbors from D

Legend



Queue: H → F → G → E

Visited: A, B, C, D



BFS Traversal

Step 10: Process node H

Legend

Complete

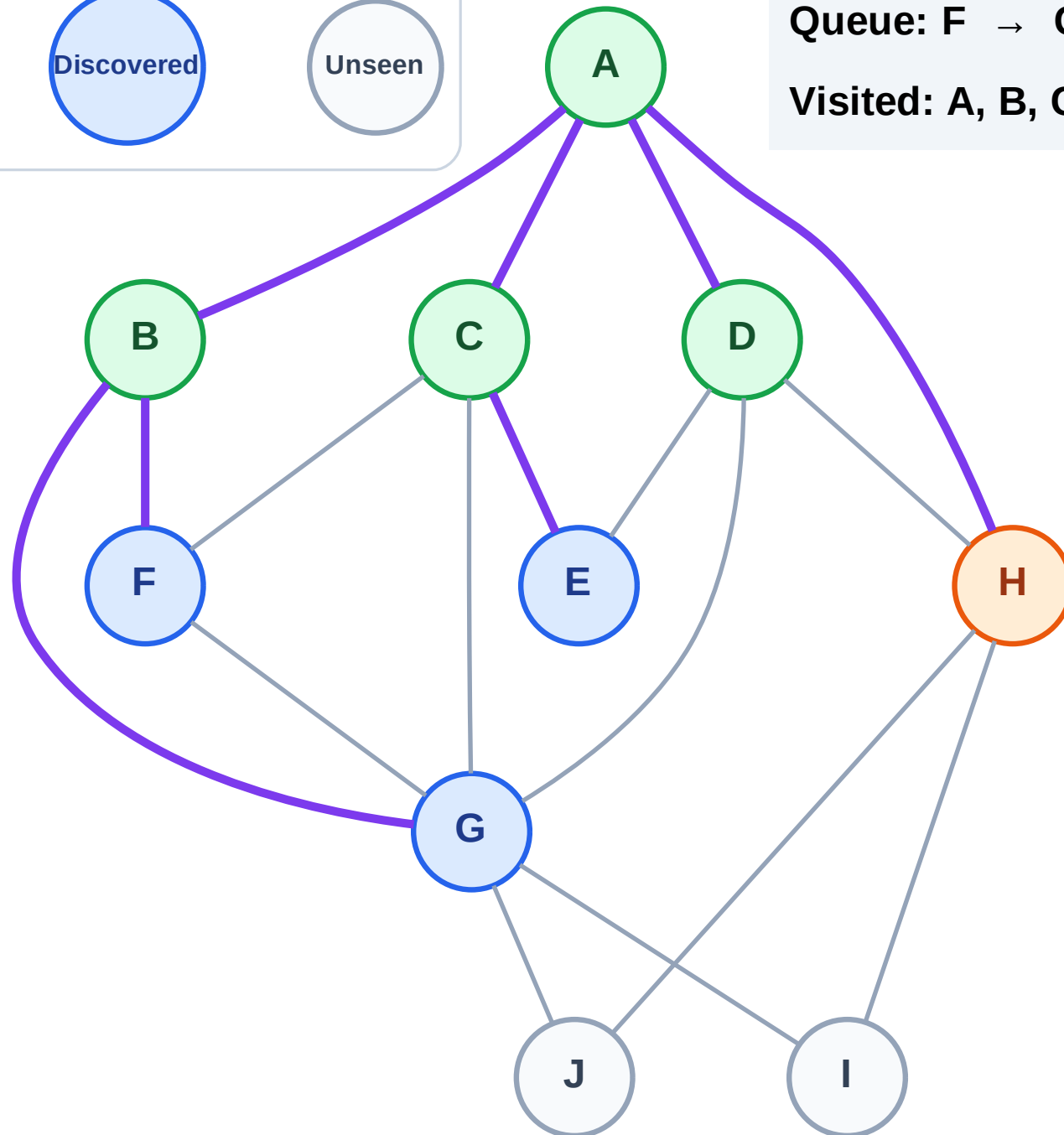
Processing

Discovered

Unseen

Queue: F → G → E

Visited: A, B, C, D



BFS Traversal

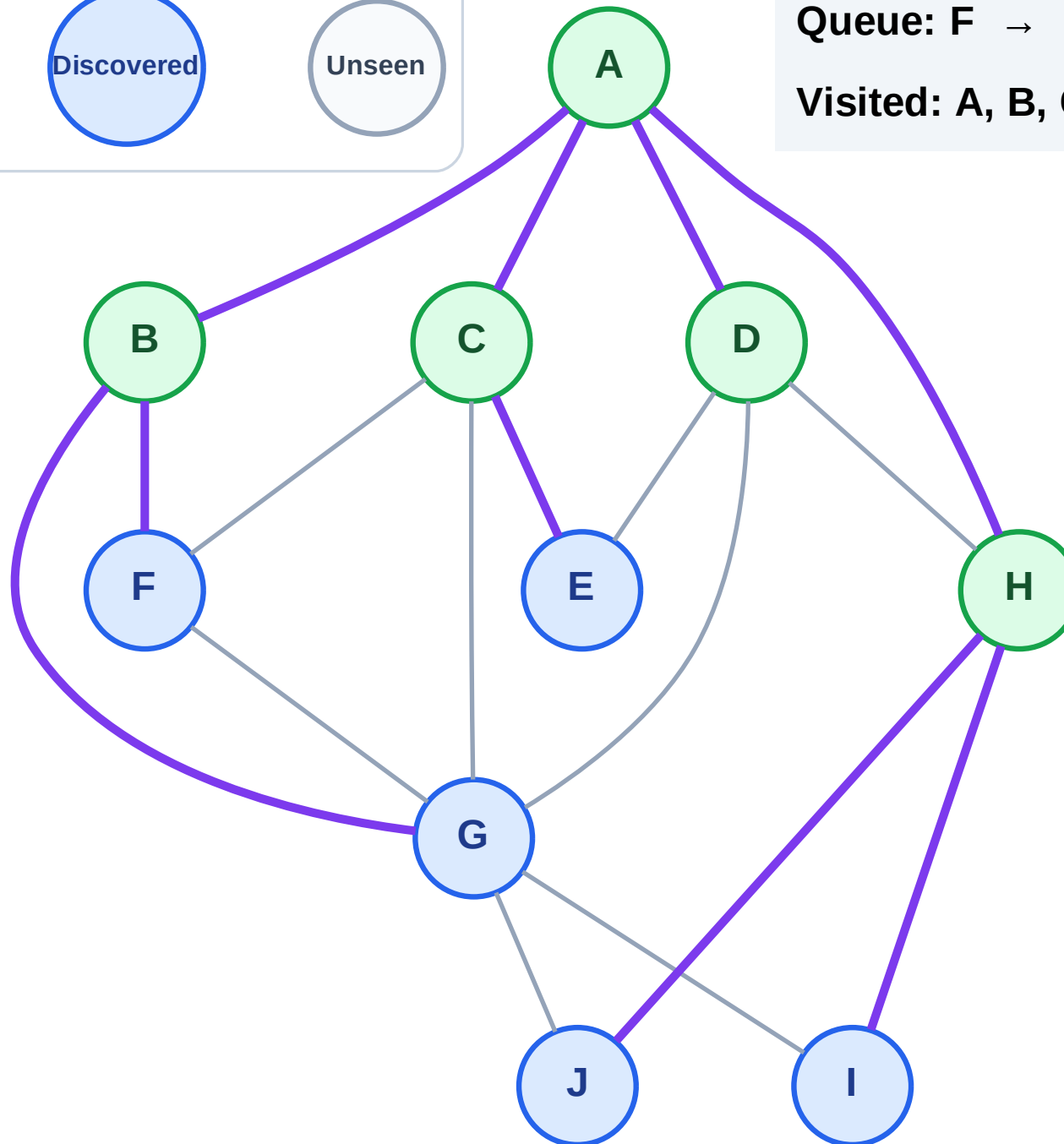
Step 11: Discover I, J from H

Legend



Queue: F → G → E → I → J

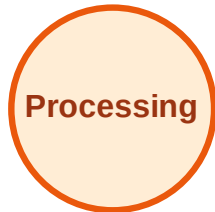
Visited: A, B, C, D, H



BFS Traversal

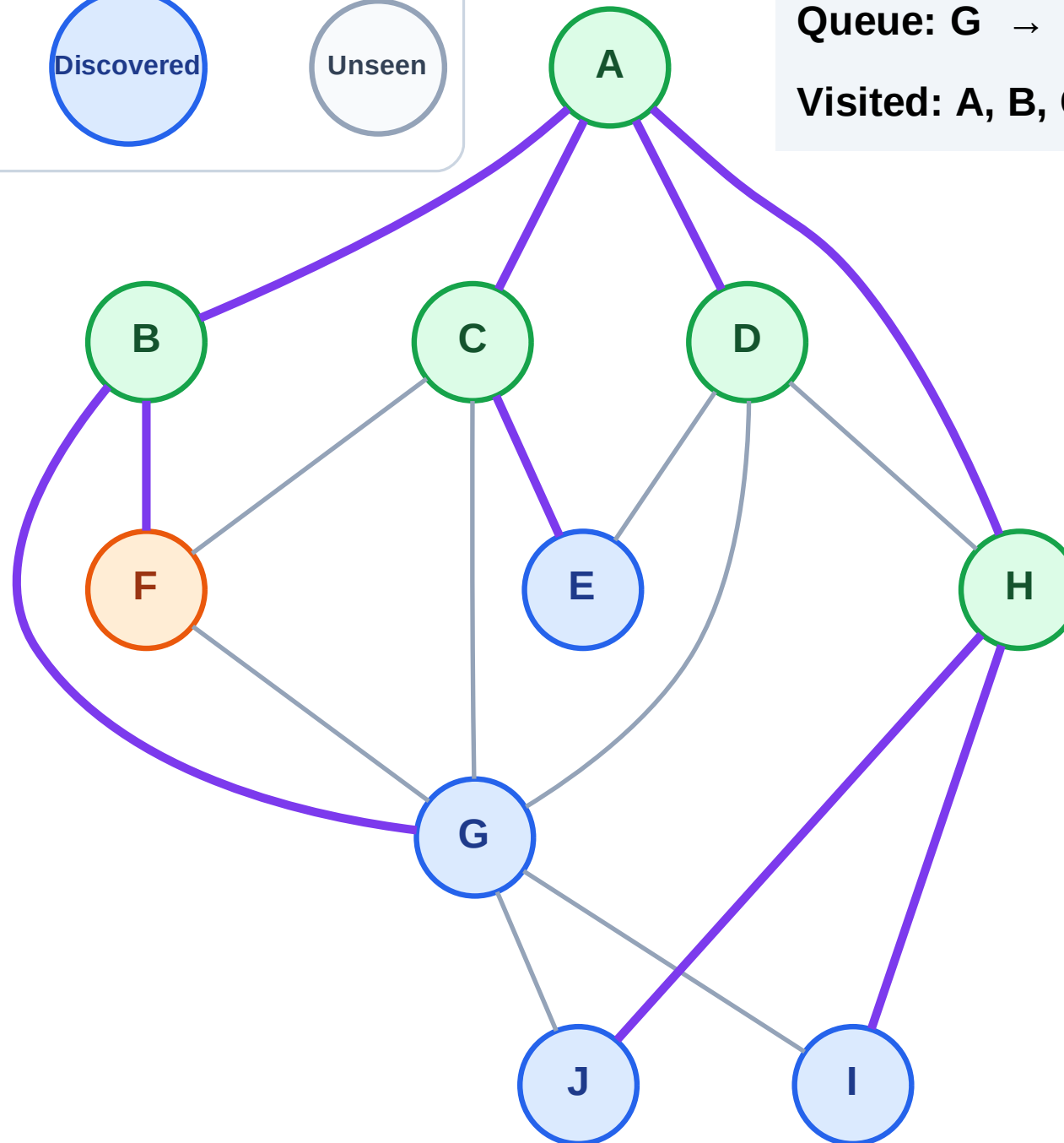
Step 12: Process node F

Legend



Queue: G → E → I → J

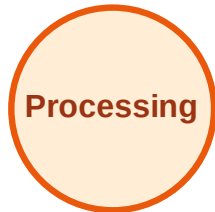
Visited: A, B, C, D, H



BFS Traversal

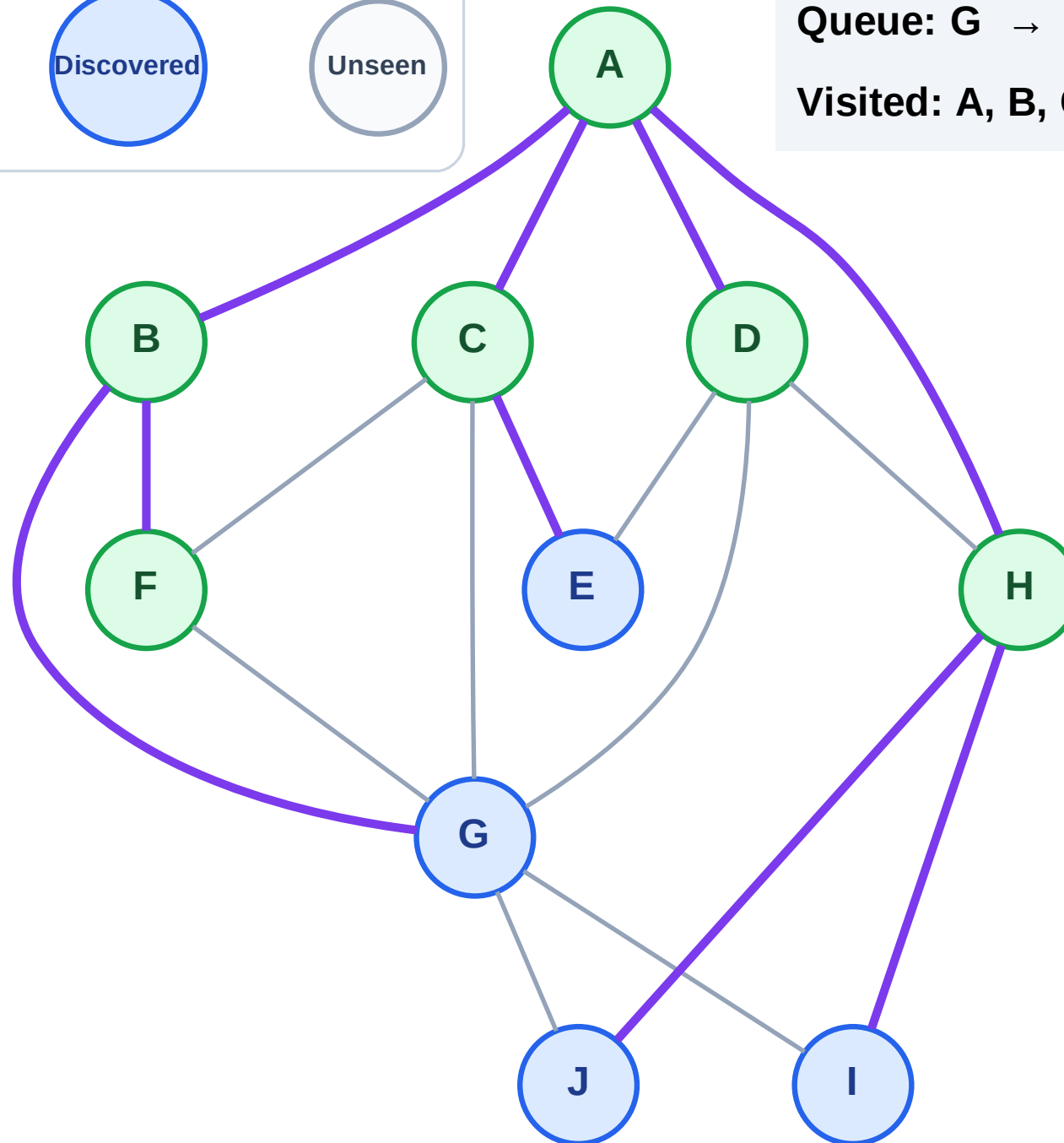
Step 13: No new neighbors from F

Legend



Queue: G → E → I → J

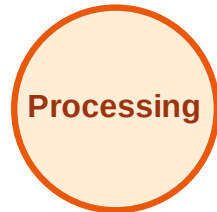
Visited: A, B, C, D, F, H



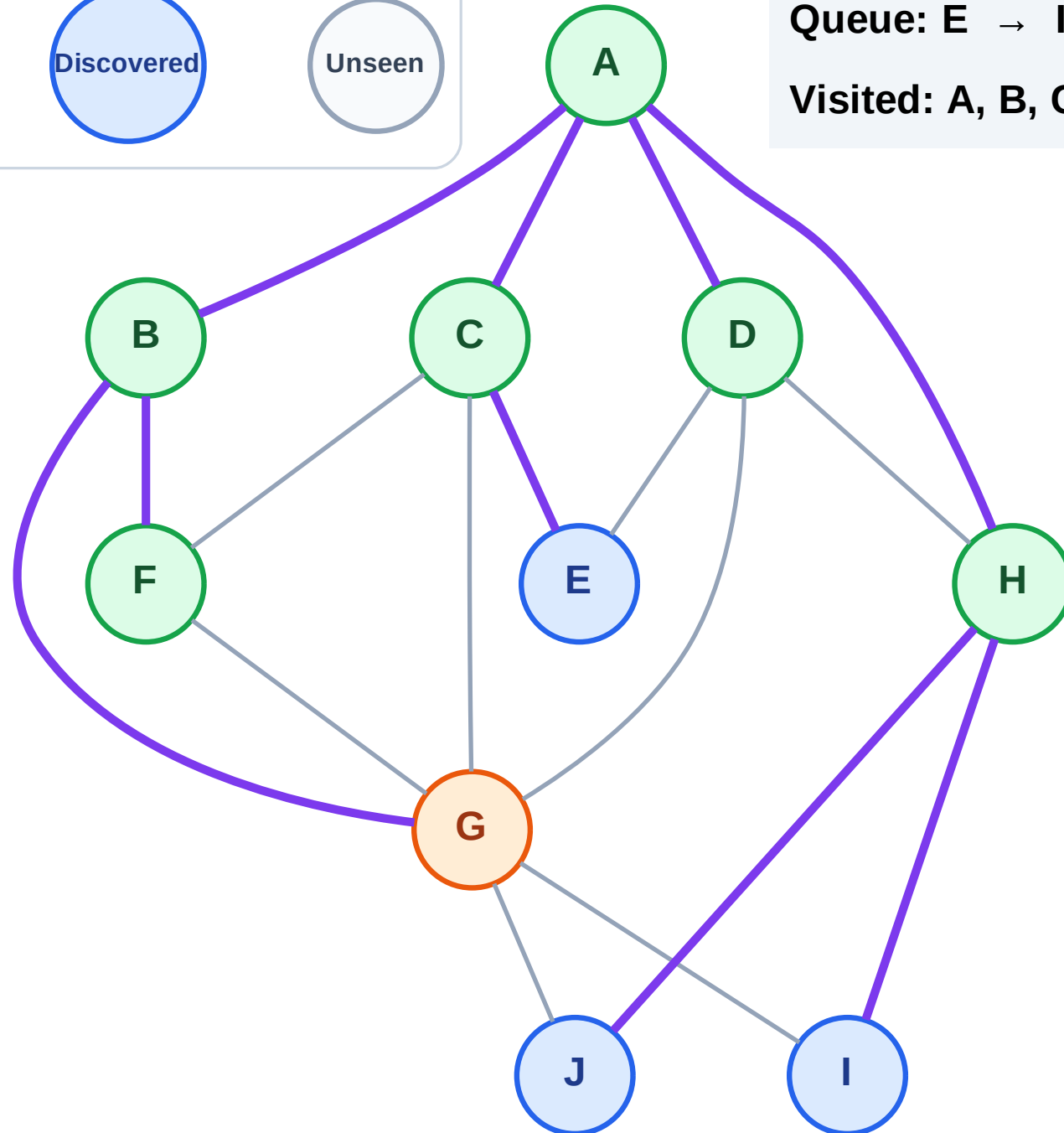
BFS Traversal

Step 14: Process node G

Legend



Queue: E → I → J
Visited: A, B, C, D, F, H



BFS Traversal

Step 15: No new neighbors from G

Legend

Complete

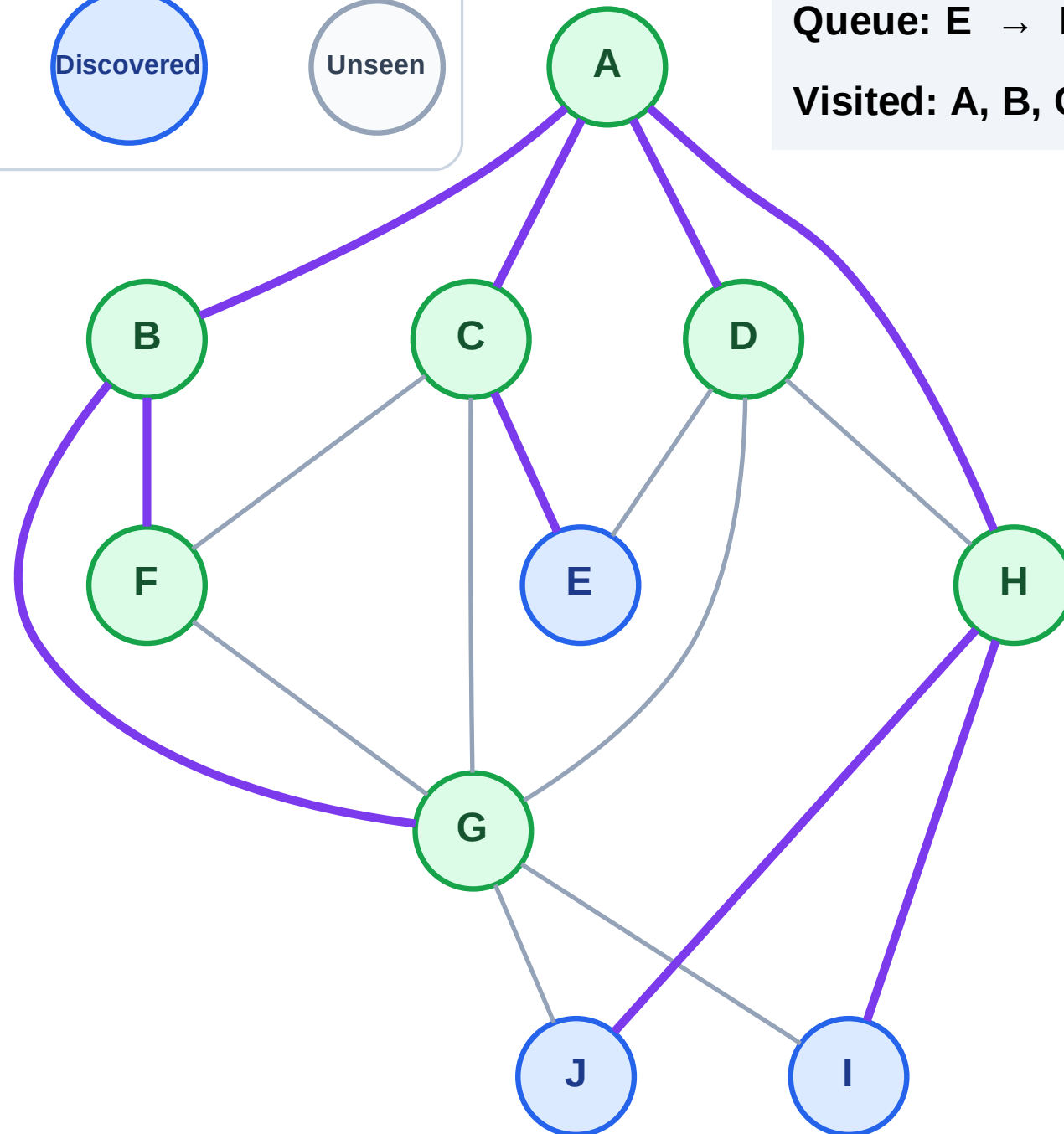
Processing

Discovered

Unseen

Queue: E → I → J

Visited: A, B, C, D, F, G, H



BFS Traversal

Step 16: Process node E

Legend

Complete

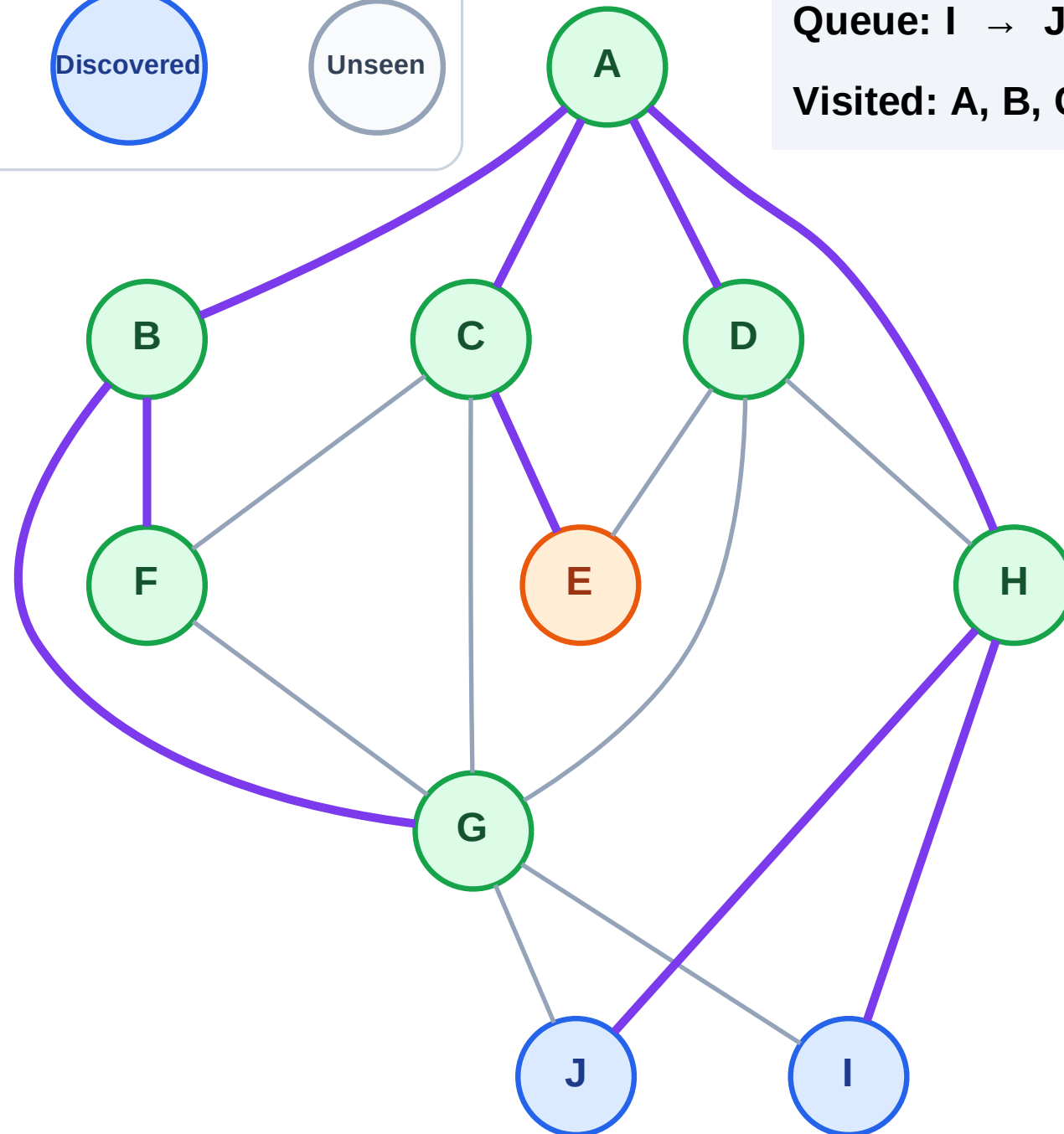
Processing

Discovered

Unseen

Queue: I → J

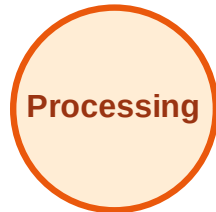
Visited: A, B, C, D, F, G, H



BFS Traversal

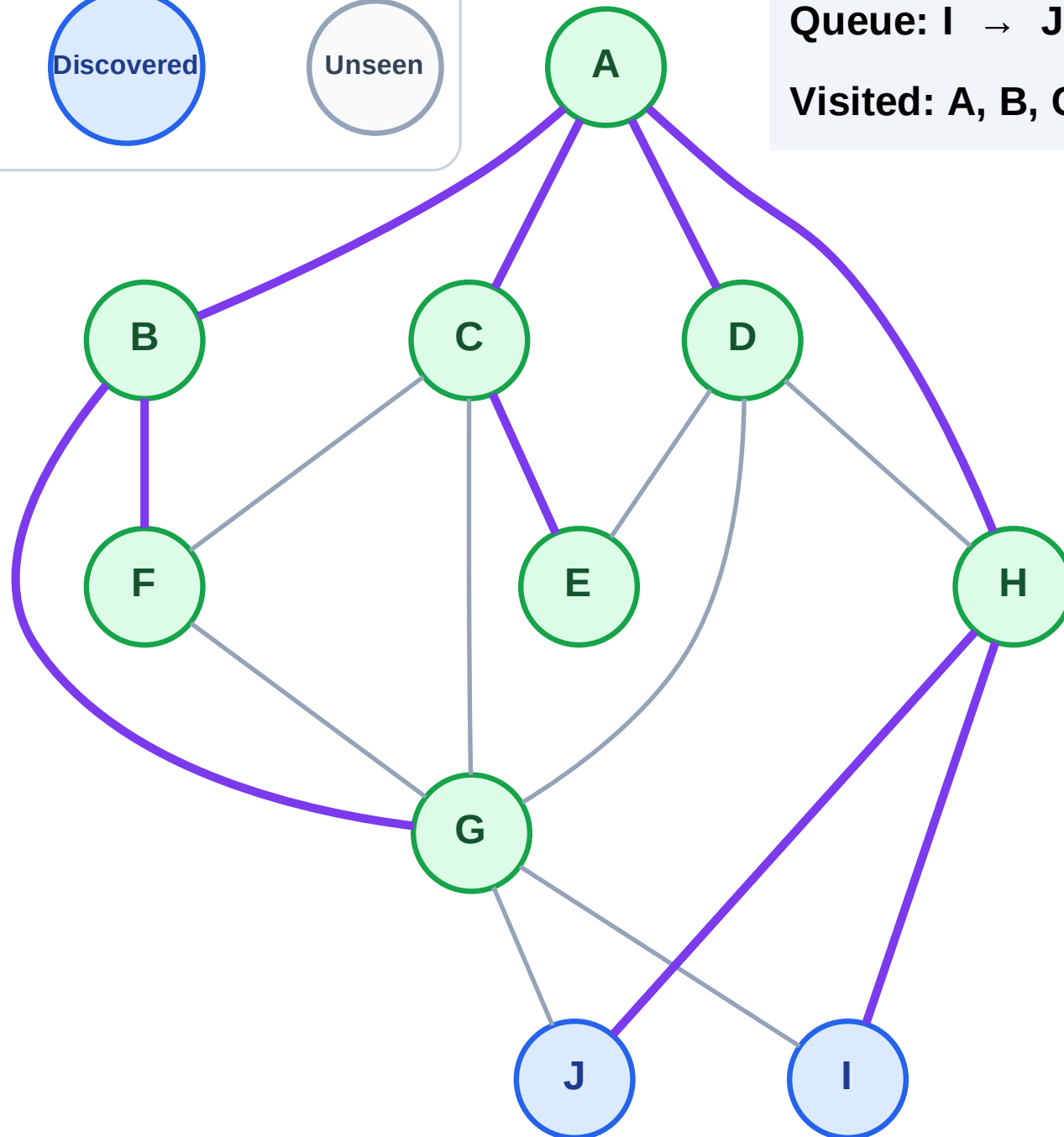
Step 17: No new neighbors from E

Legend



Queue: I → J

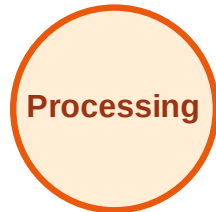
Visited: A, B, C, D, E, F, G, H



BFS Traversal

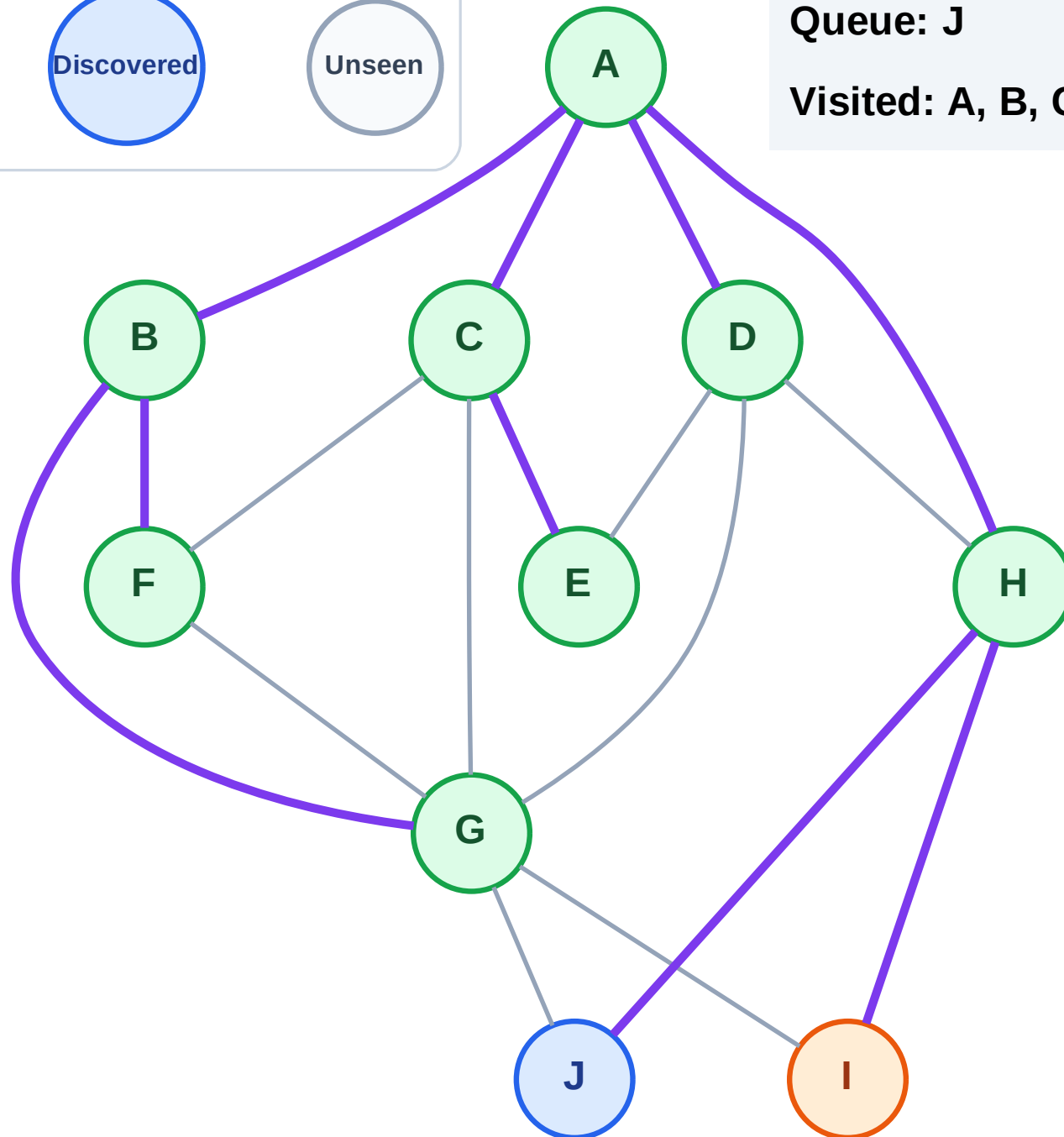
Step 18: Process node I

Legend



Queue: J

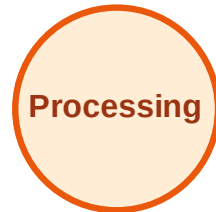
Visited: A, B, C, D, E, F, G, H



BFS Traversal

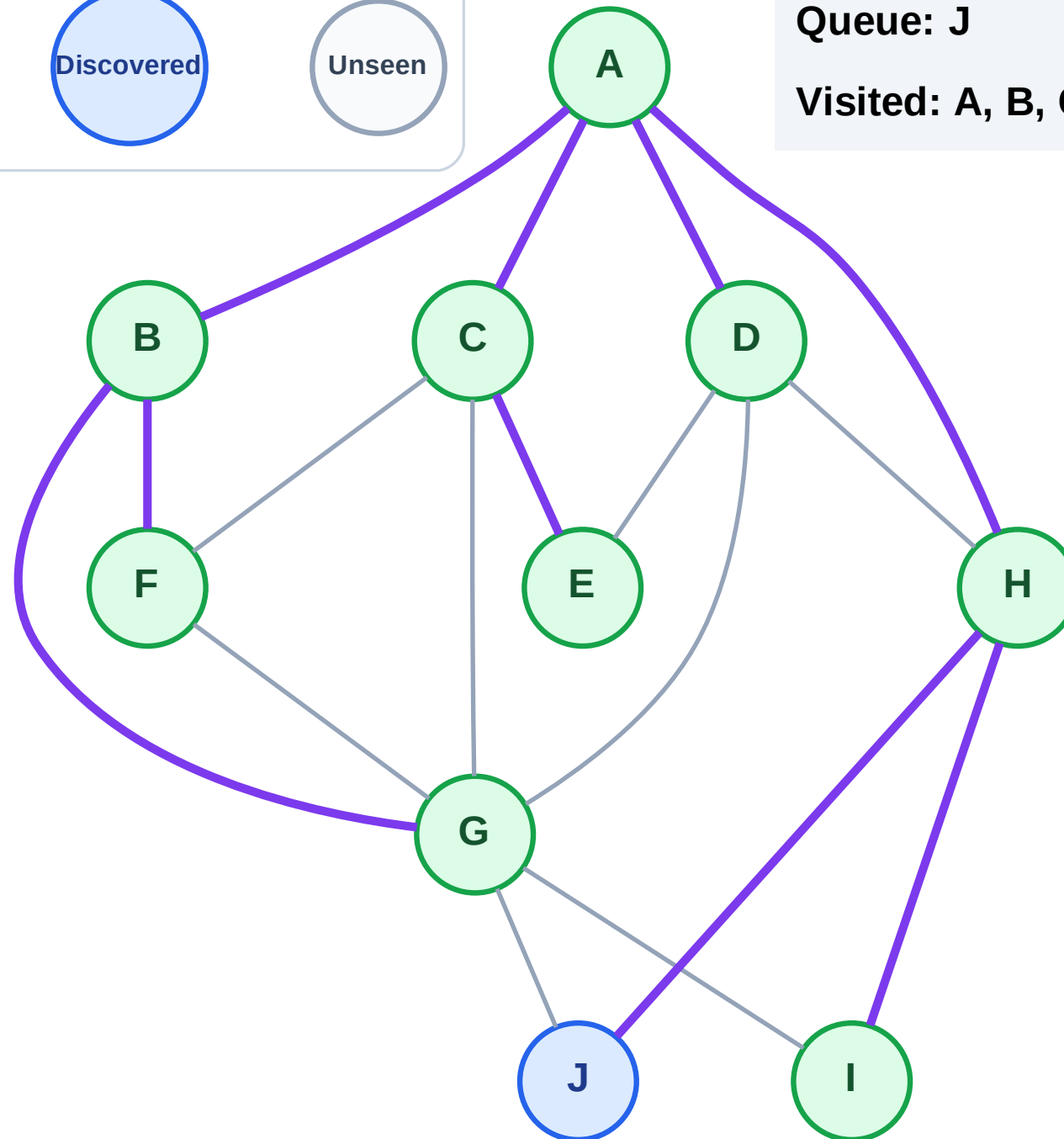
Step 19: No new neighbors from I

Legend



Queue: J

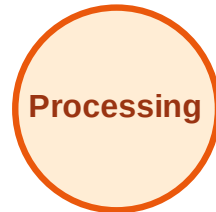
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

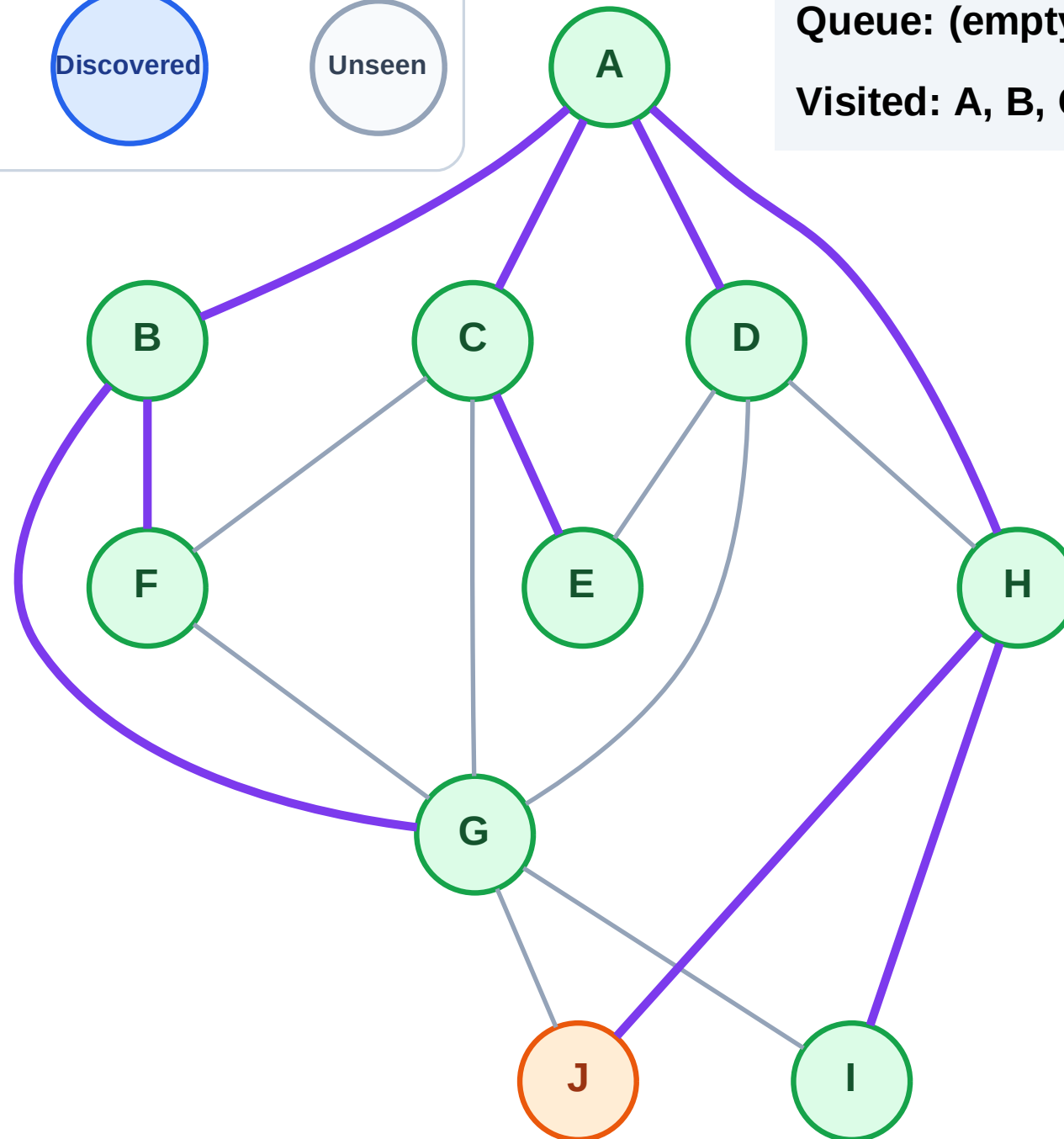
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

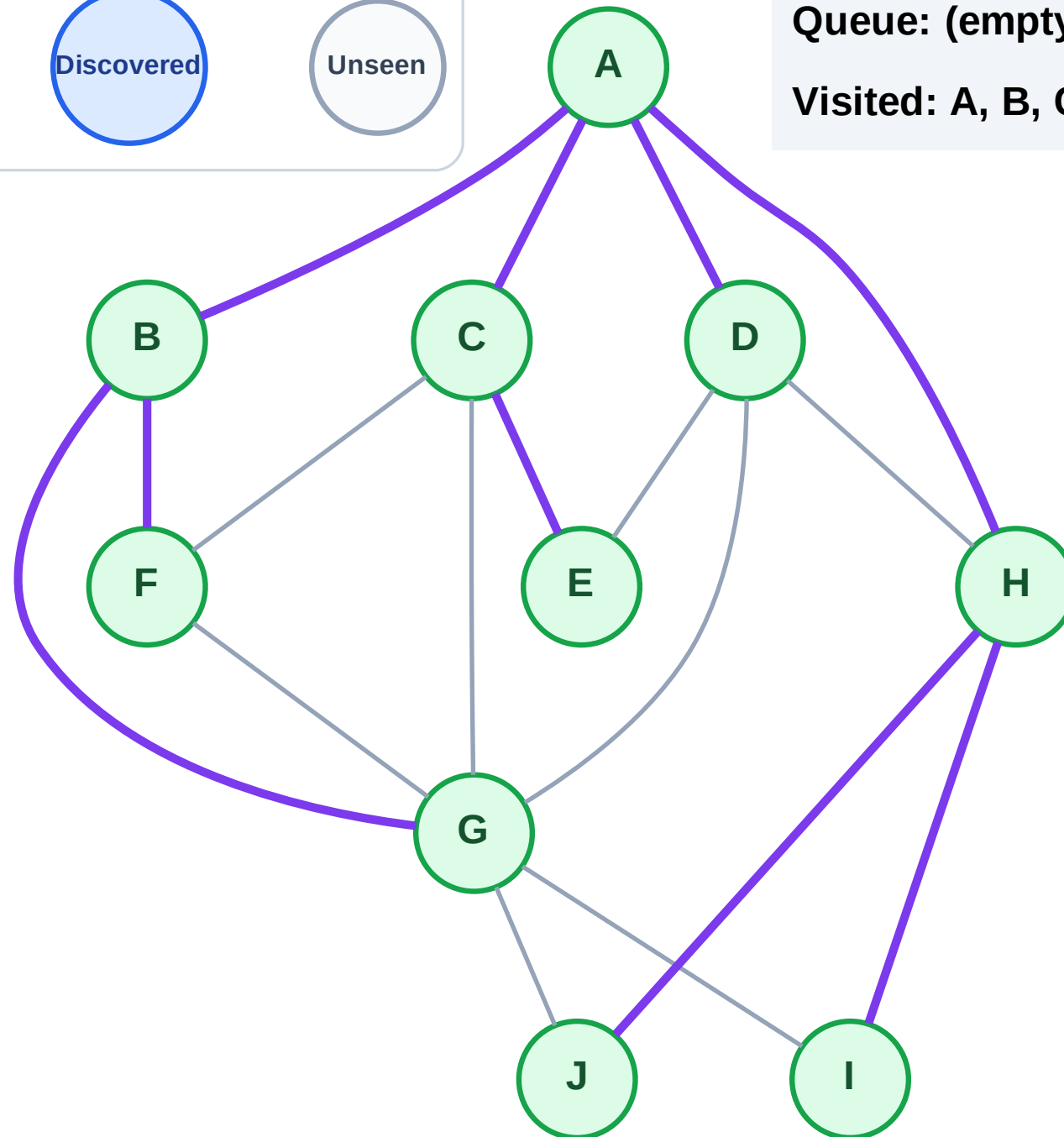
Step 21: No new neighbors from J

Legend



Queue: (empty)

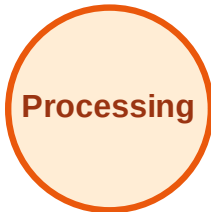
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

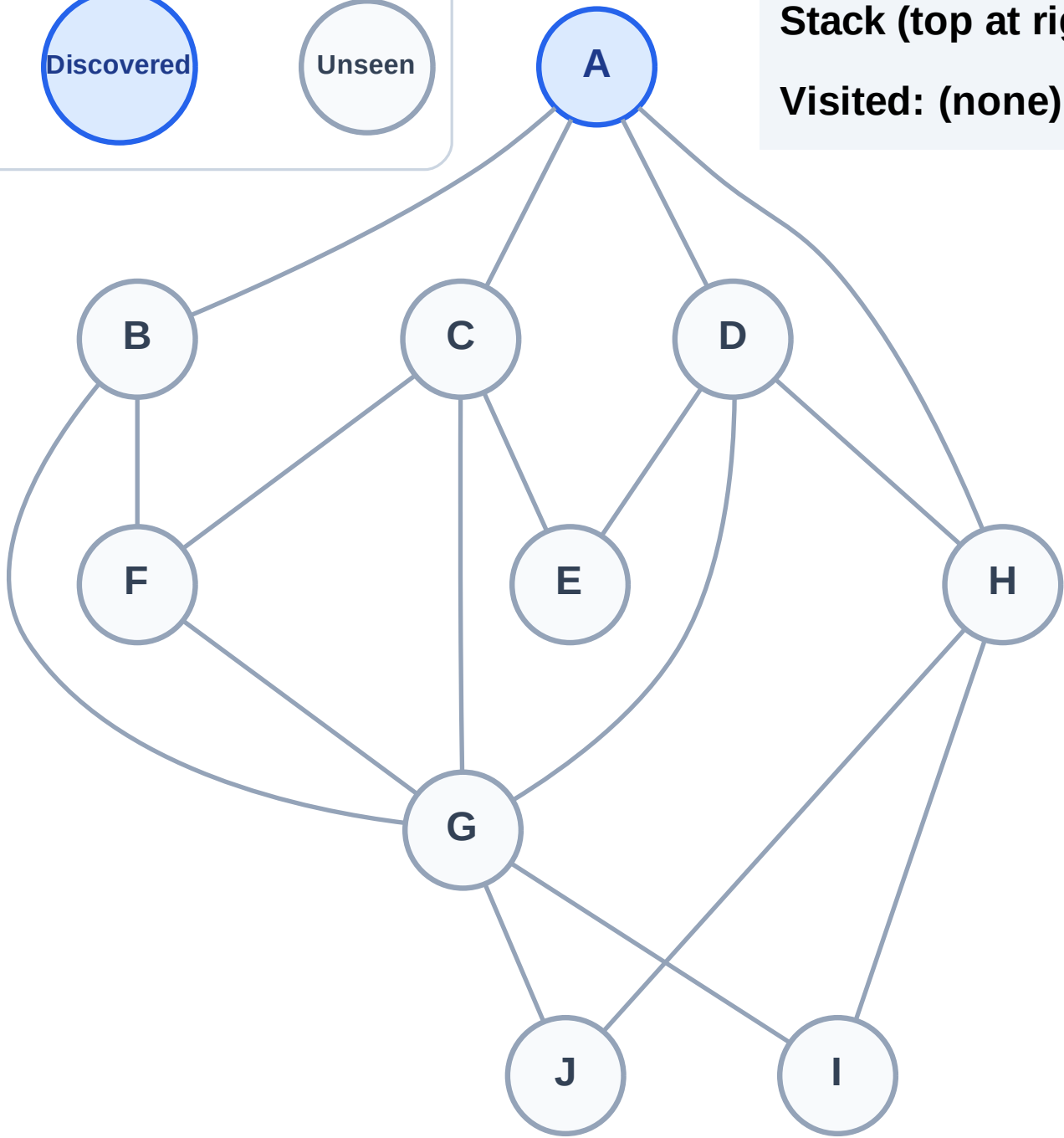
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

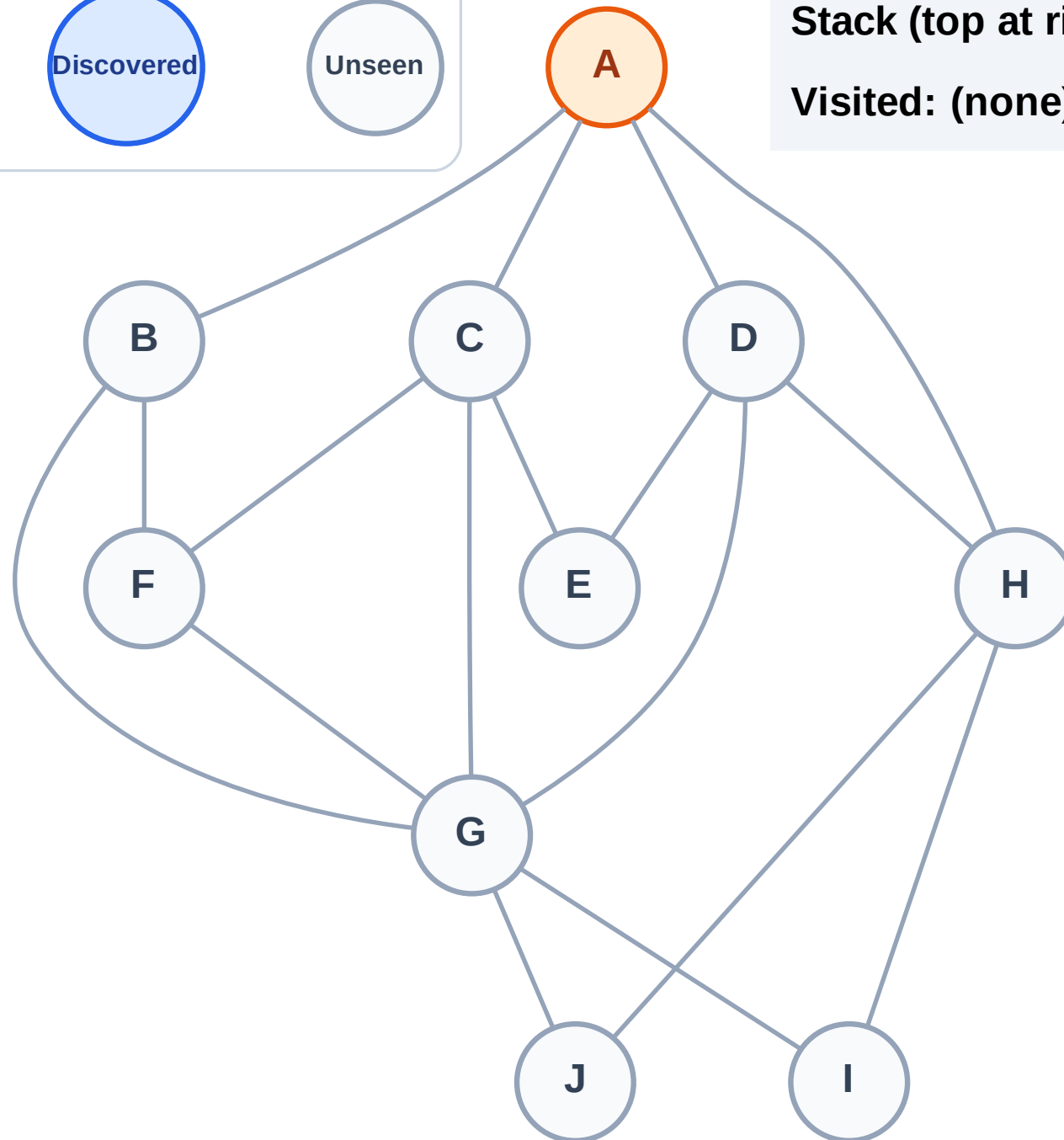
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover H, D, C, B from A

Legend

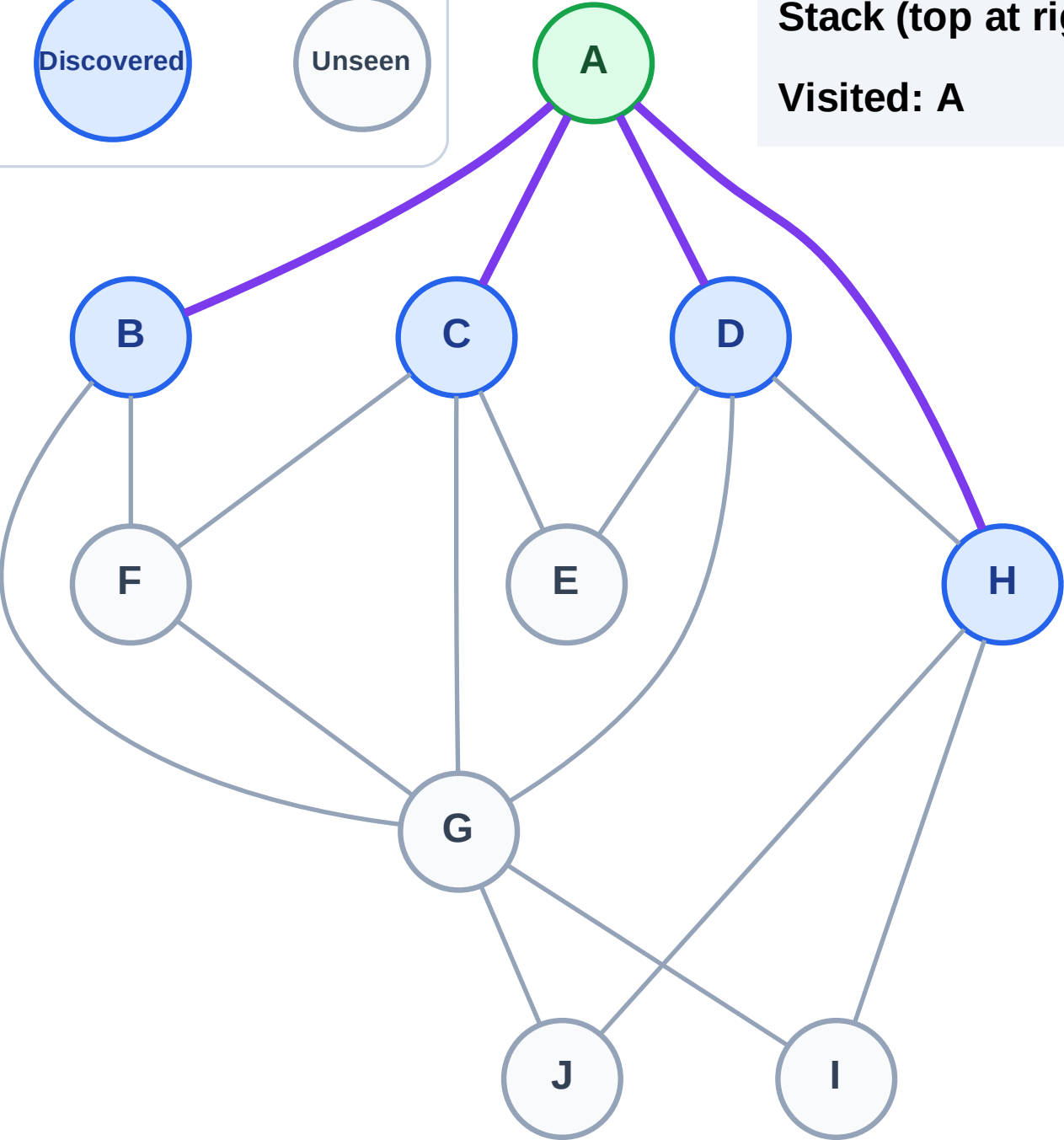
Complete

Processing

Discovered

Unseen

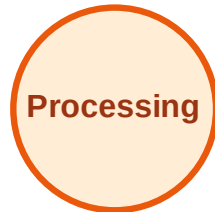
Stack (top at right): H | D | C | B
Visited: A



DFS Traversal

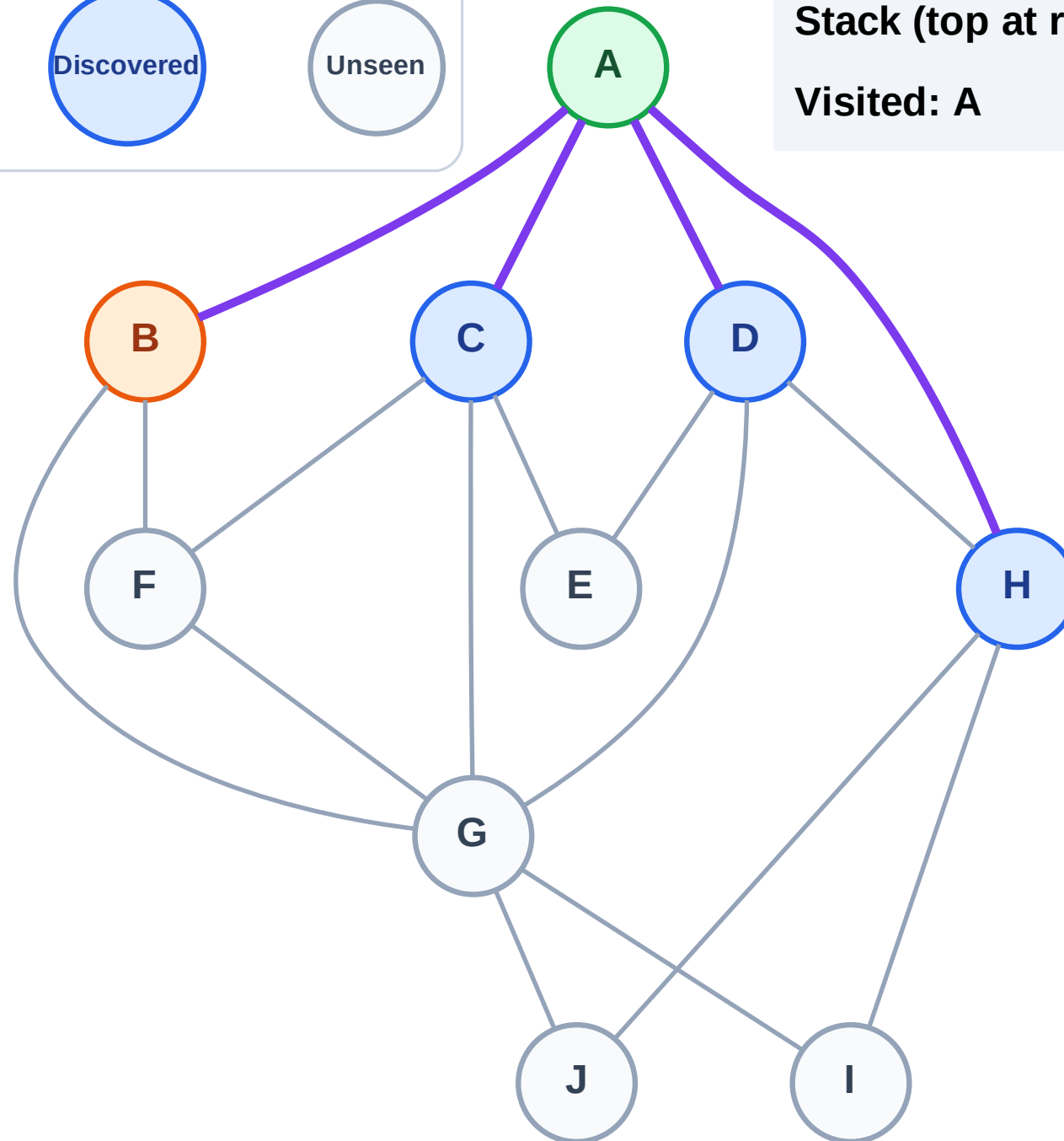
Step 4: Process node B

Legend



Stack (top at right): H | D | C

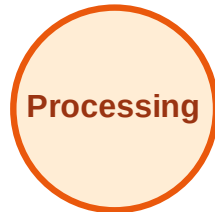
Visited: A



DFS Traversal

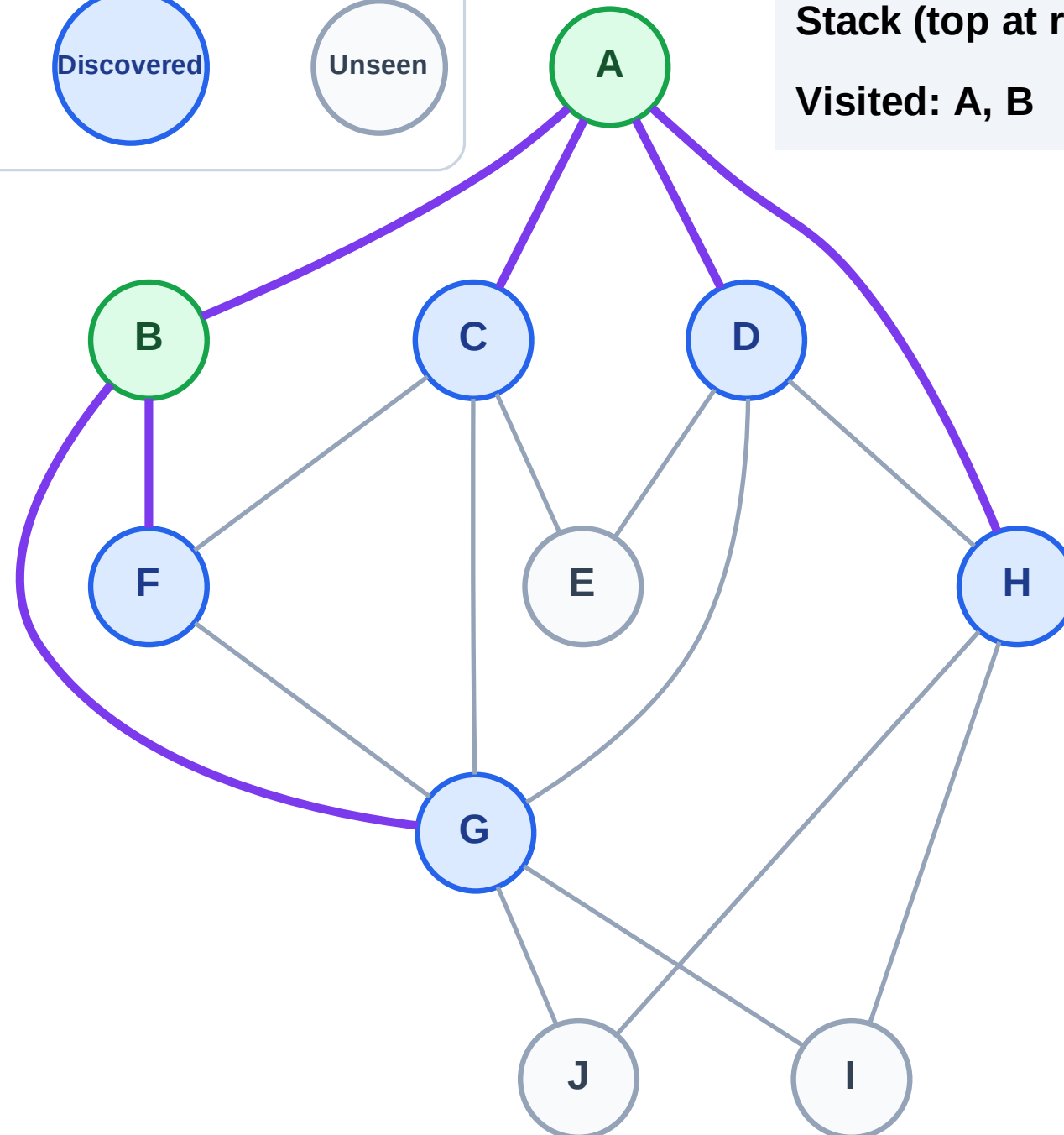
Step 5: Discover G, F from B

Legend



Stack (top at right): H | D | C | G | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

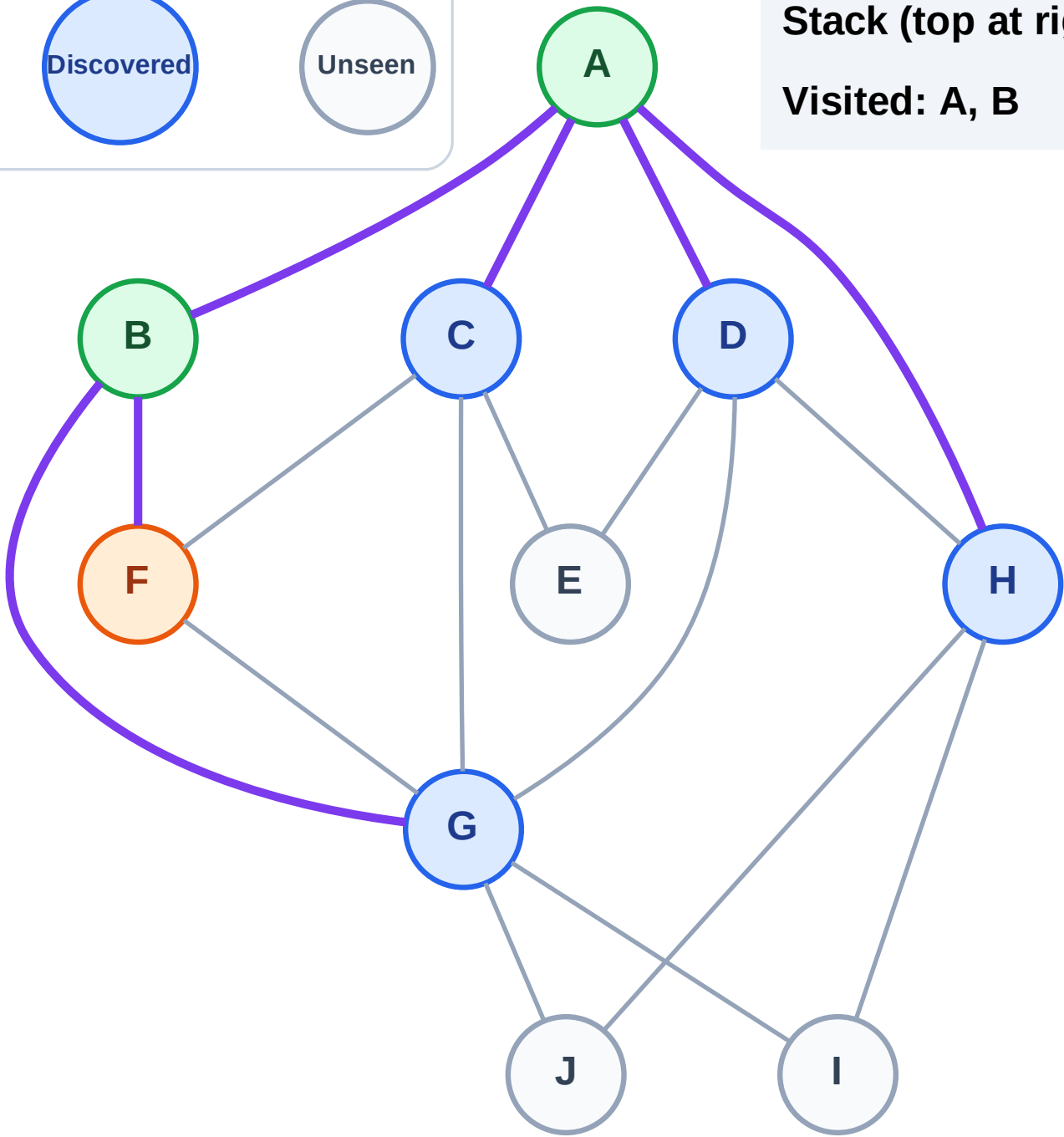
Processing

Discovered

Unseen

Stack (top at right): H | D | C | G

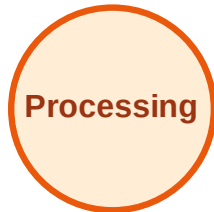
Visited: A, B



DFS Traversal

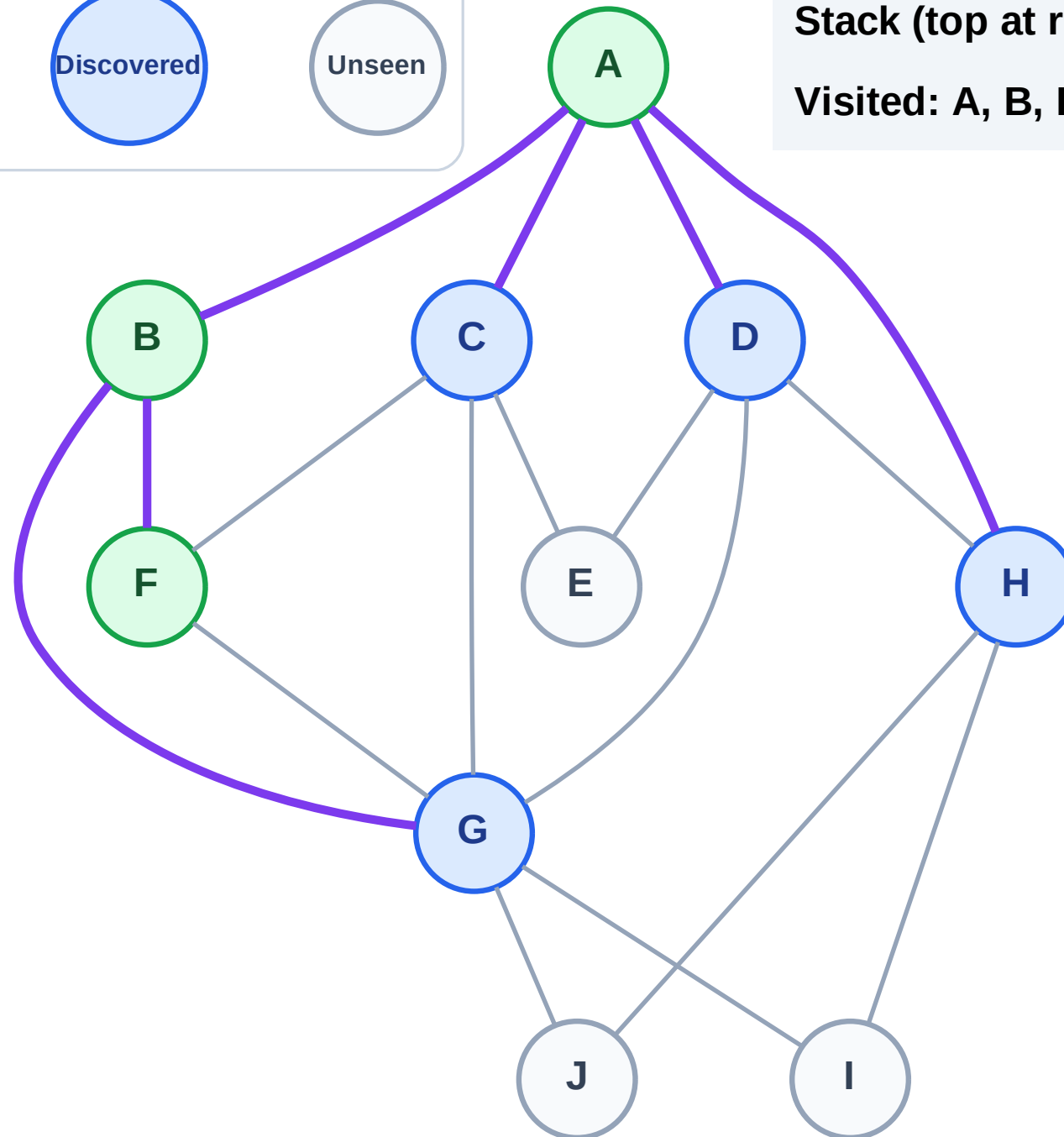
Step 7: No new neighbors from F

Legend



Stack (top at right): H | D | C | G

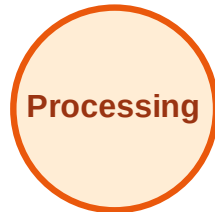
Visited: A, B, F



DFS Traversal

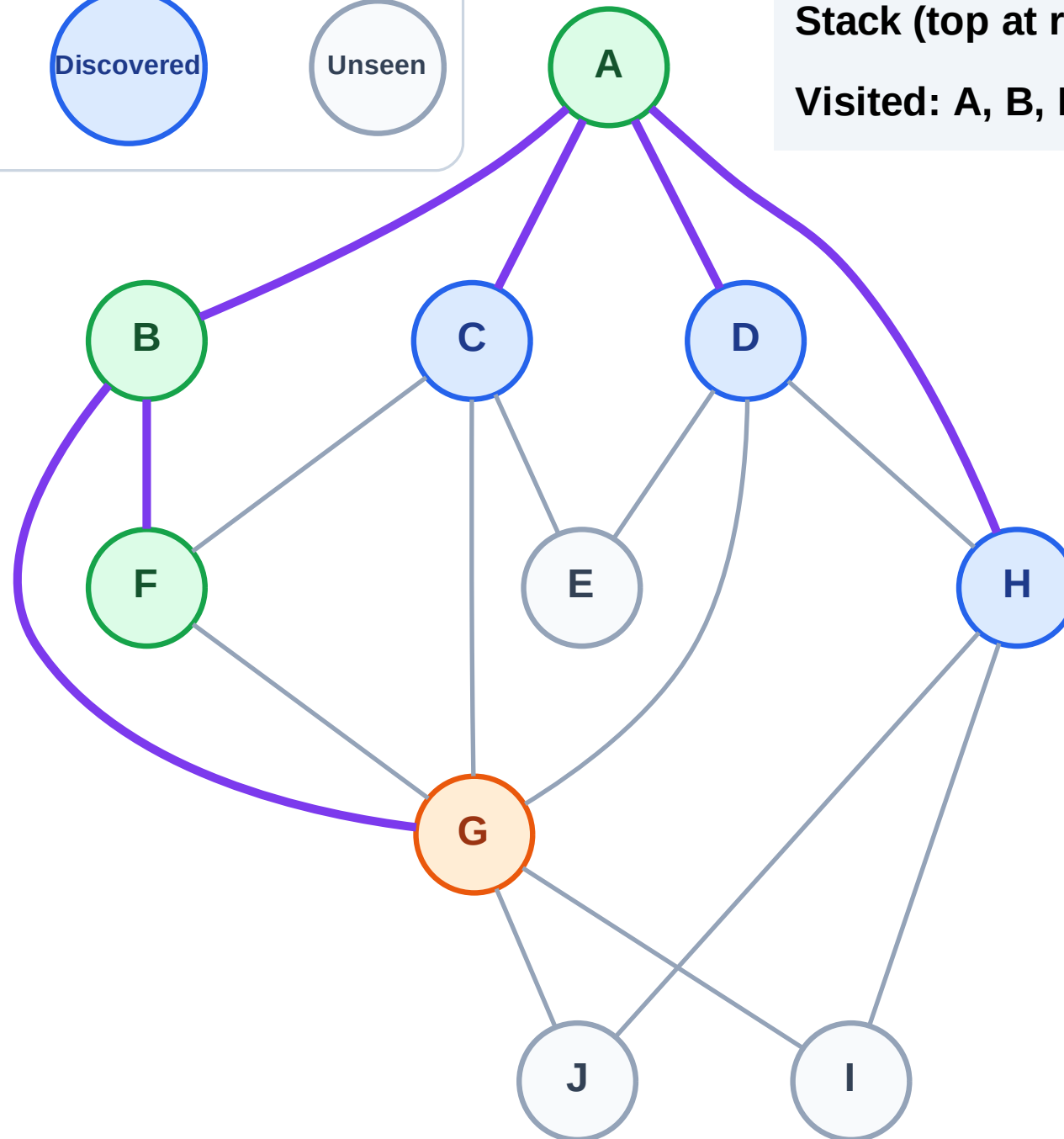
Step 8: Process node G

Legend



Stack (top at right): H | D | C

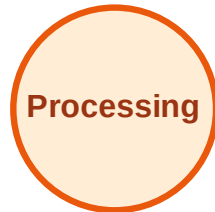
Visited: A, B, F



DFS Traversal

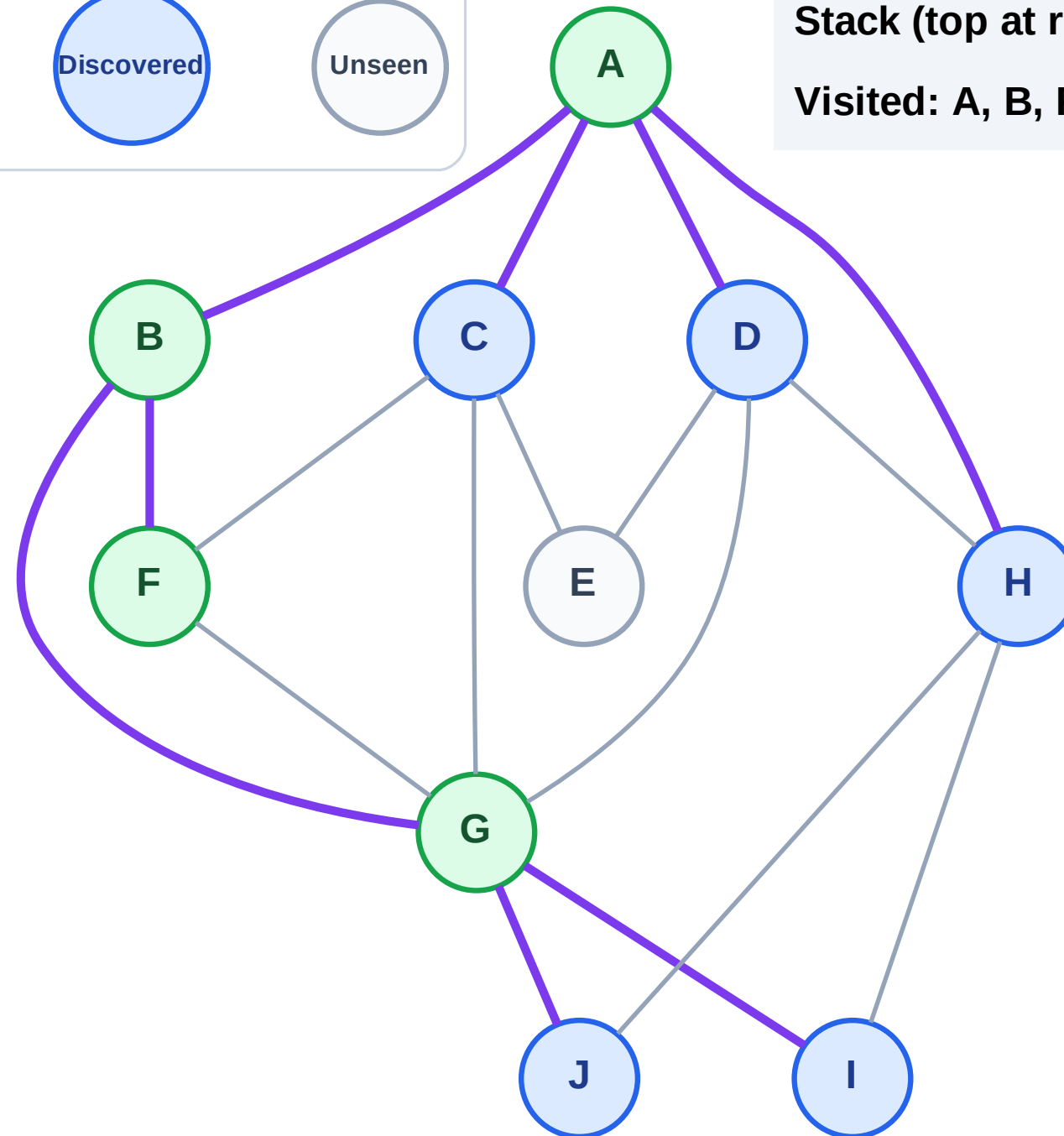
Step 9: Discover J, I from G

Legend



Stack (top at right): H | D | C | J | I

Visited: A, B, F, G



DFS Traversal

Step 10: Process node I

Legend

Complete

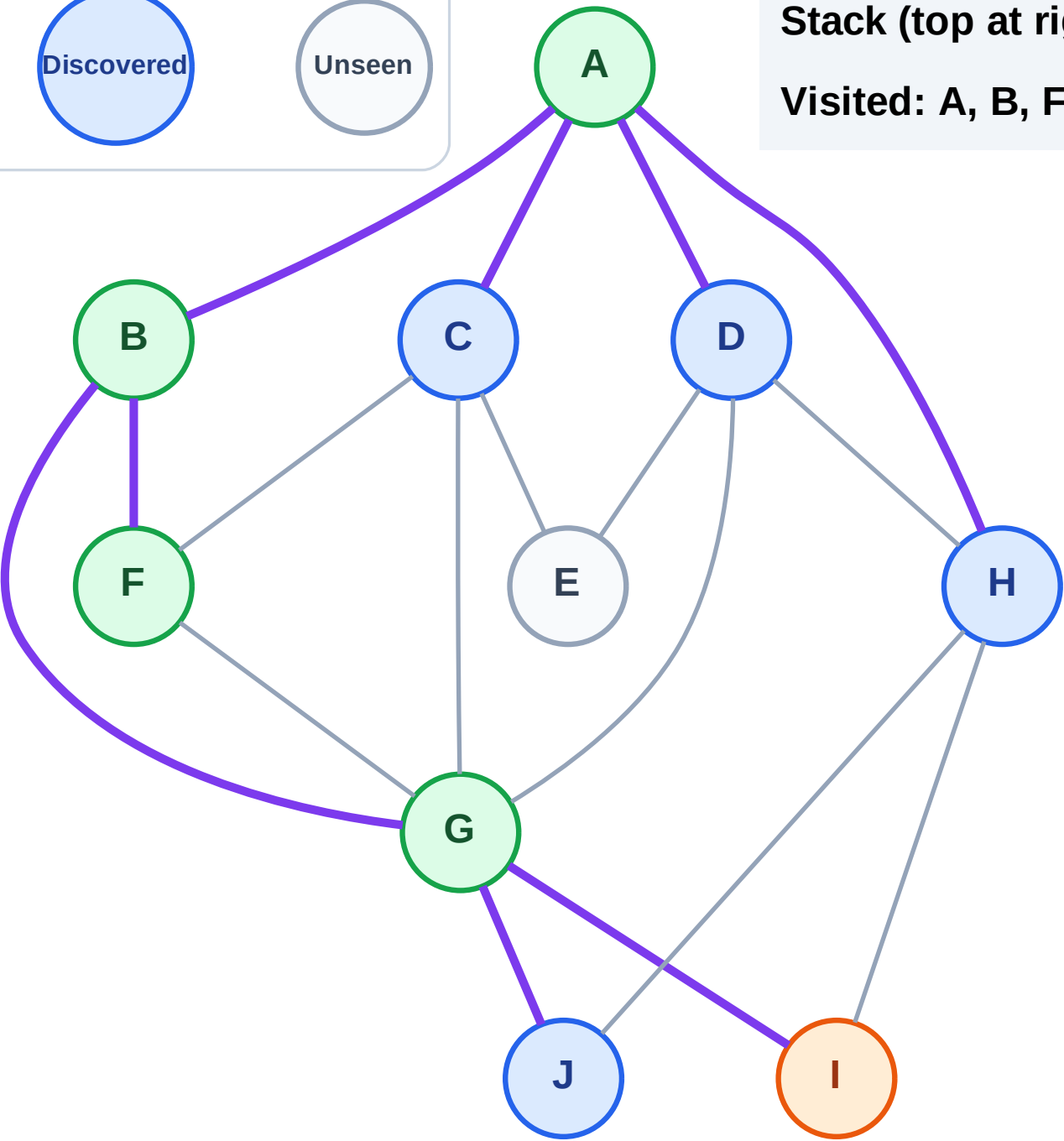
Processing

Discovered

Unseen

Stack (top at right): H | D | C | J

Visited: A, B, F, G



DFS Traversal

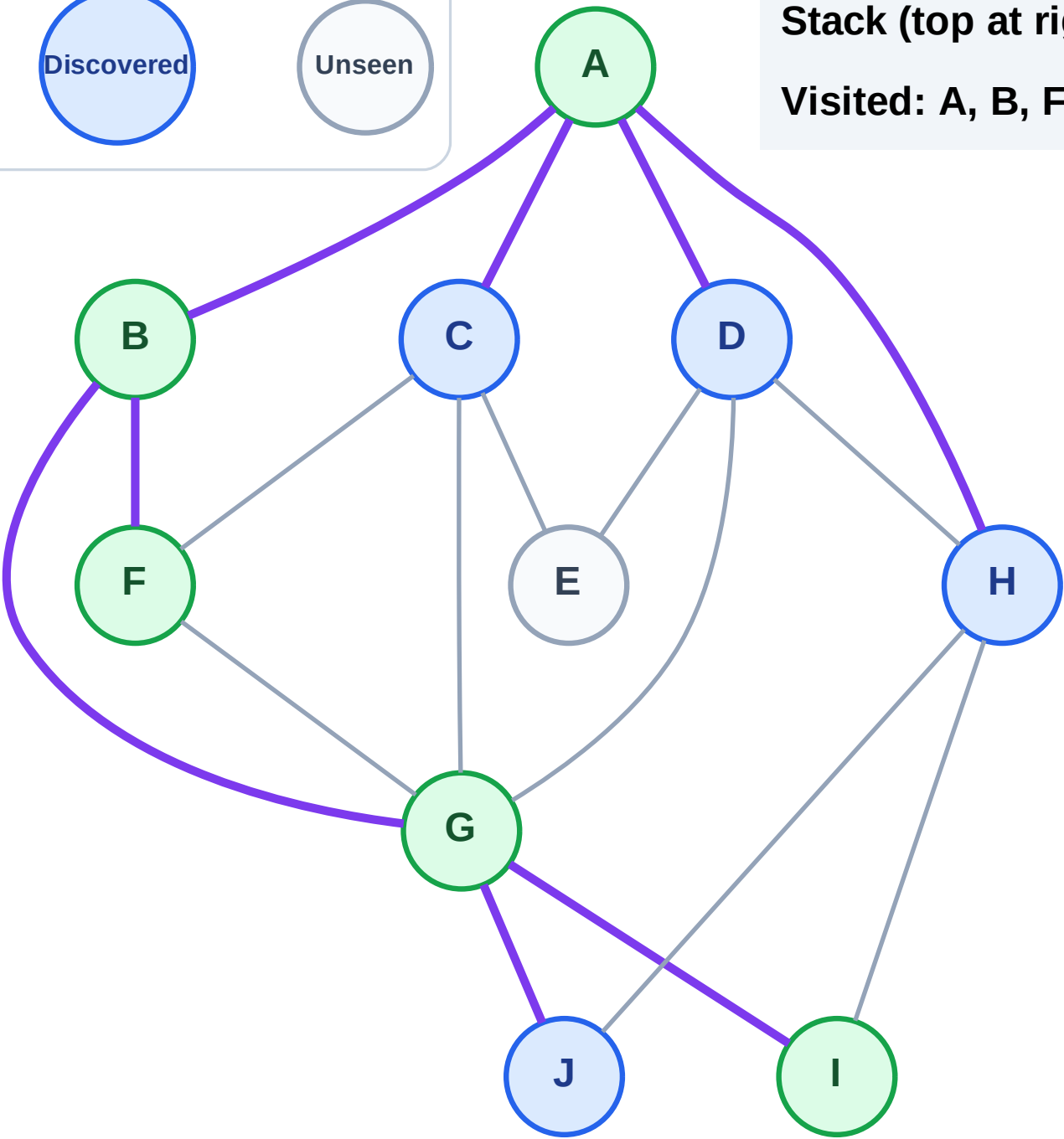
Step 11: No new neighbors from I

Legend



Stack (top at right): H | D | C | J

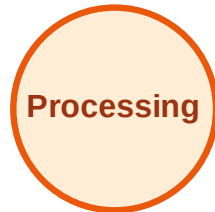
Visited: A, B, F, G, I



DFS Traversal

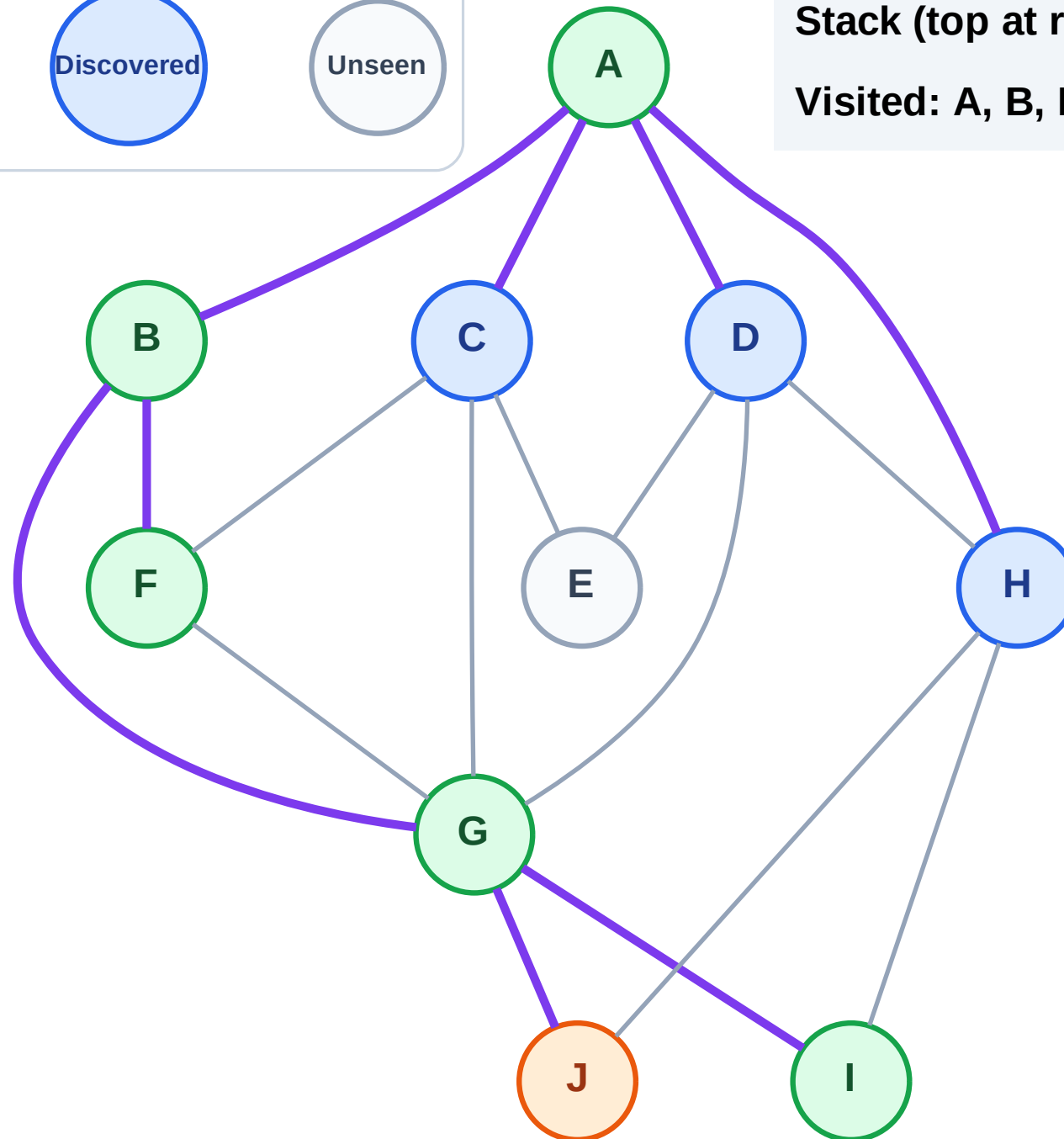
Step 12: Process node J

Legend



Stack (top at right): H | D | C

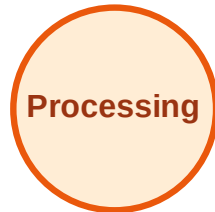
Visited: A, B, F, G, I



DFS Traversal

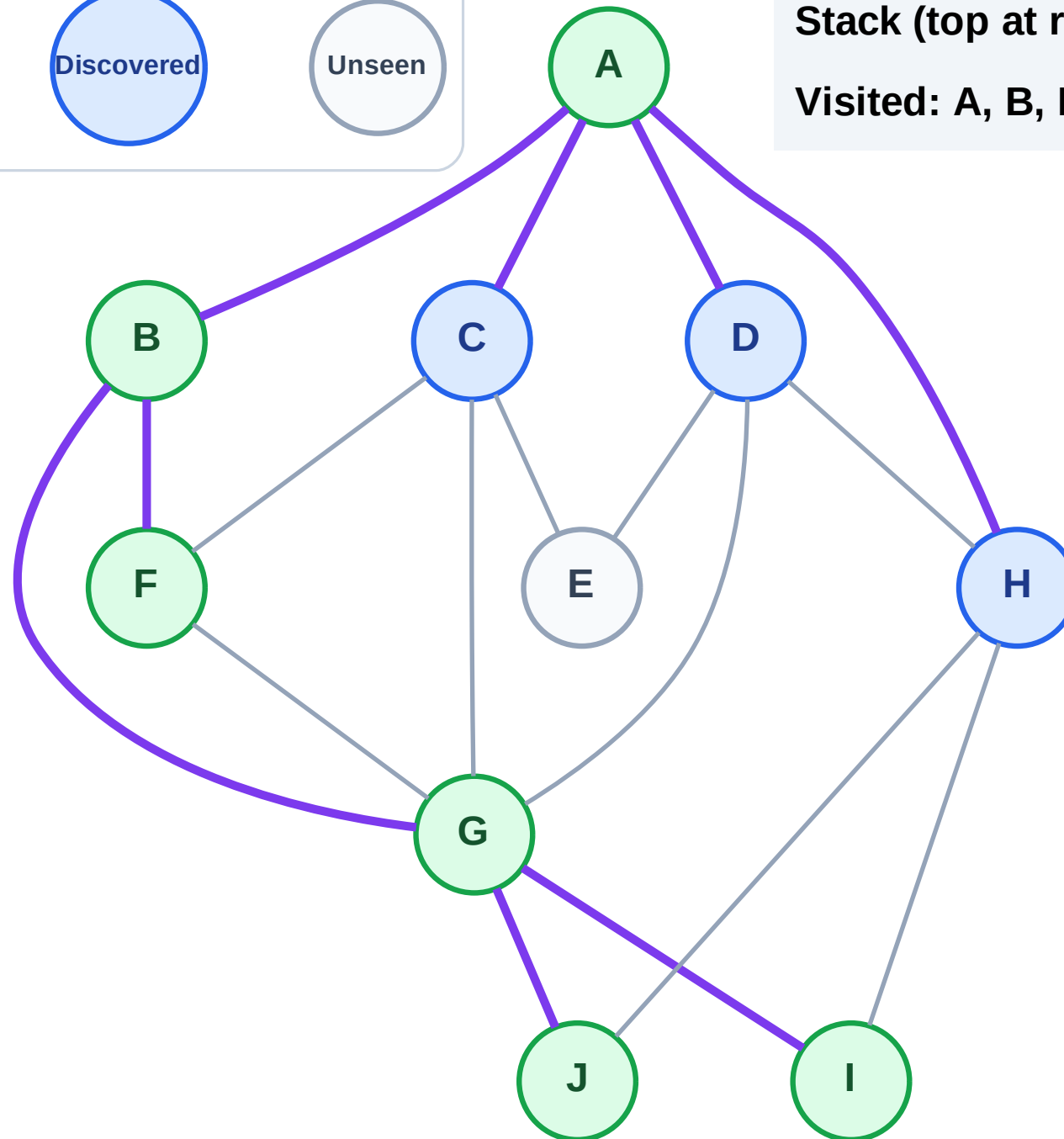
Step 13: No new neighbors from J

Legend



Stack (top at right): H | D | C

Visited: A, B, F, G, I, J



DFS Traversal

Step 14: Process node C

Legend

Complete

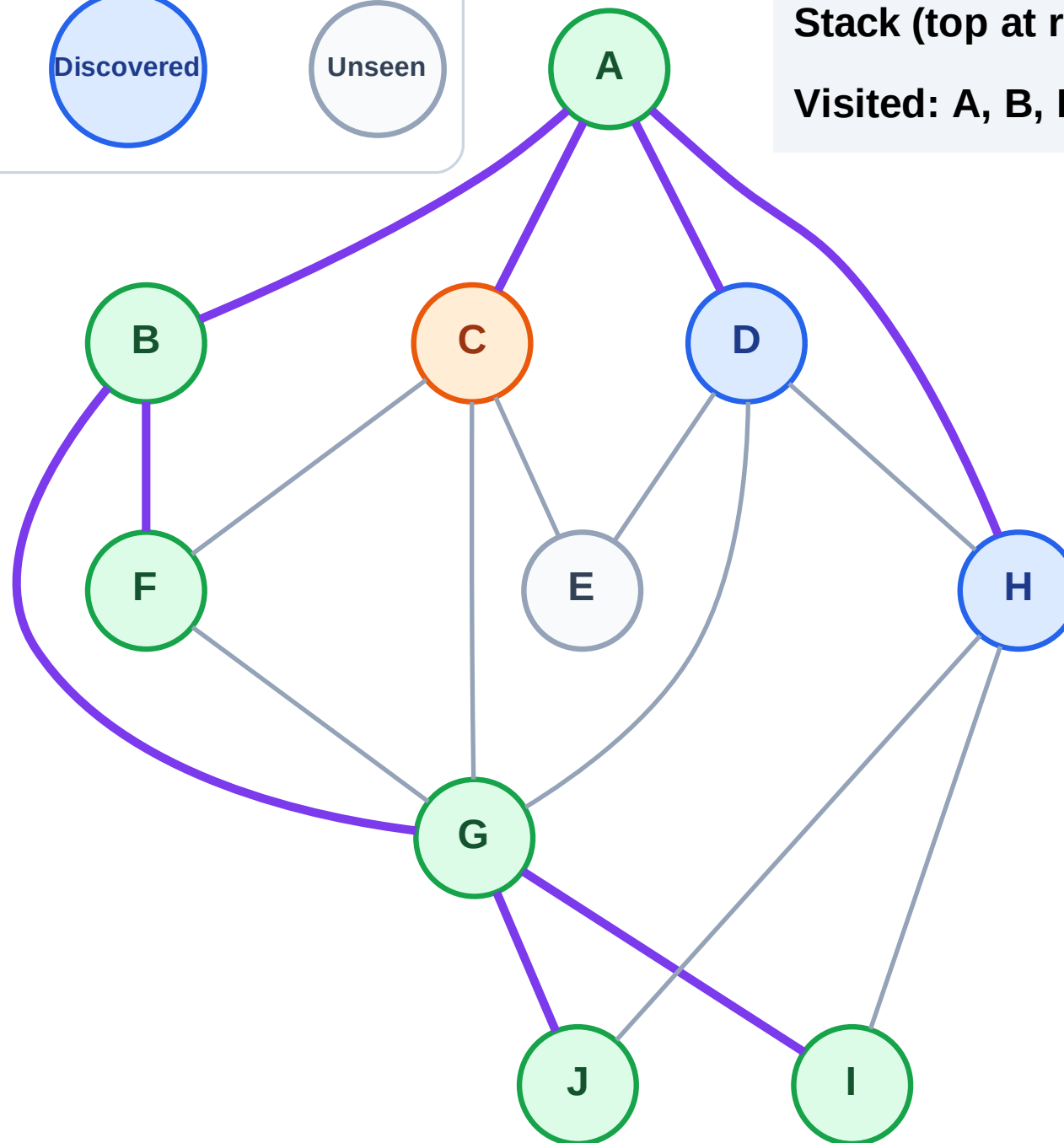
Processing

Discovered

Unseen

Stack (top at right): H | D

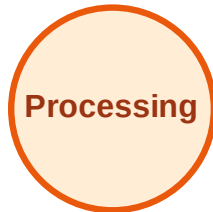
Visited: A, B, F, G, I, J



DFS Traversal

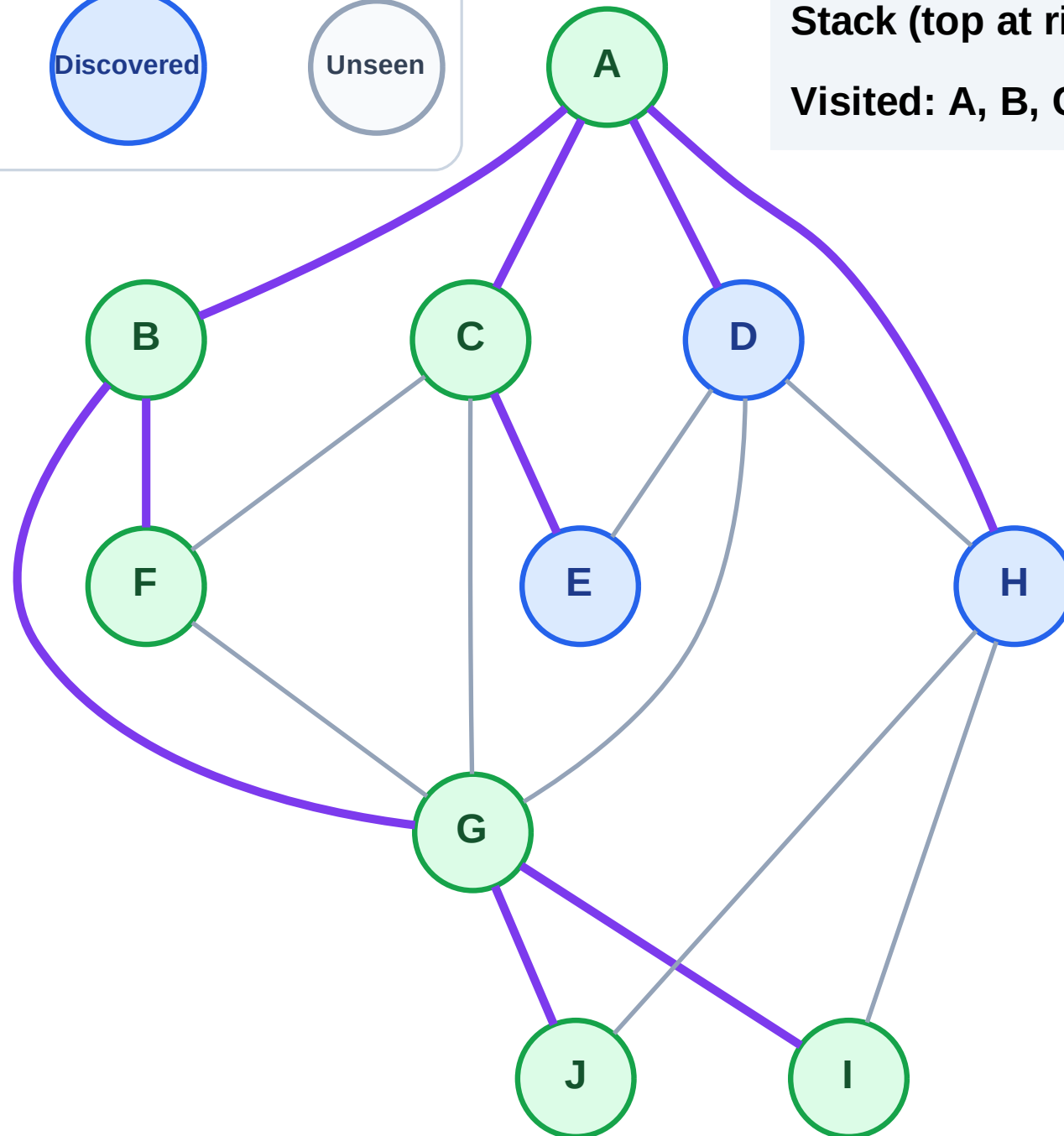
Step 15: Discover E from C

Legend



Stack (top at right): H | D | E

Visited: A, B, C, F, G, I, J



DFS Traversal

Step 16: Process node E

Legend

Complete

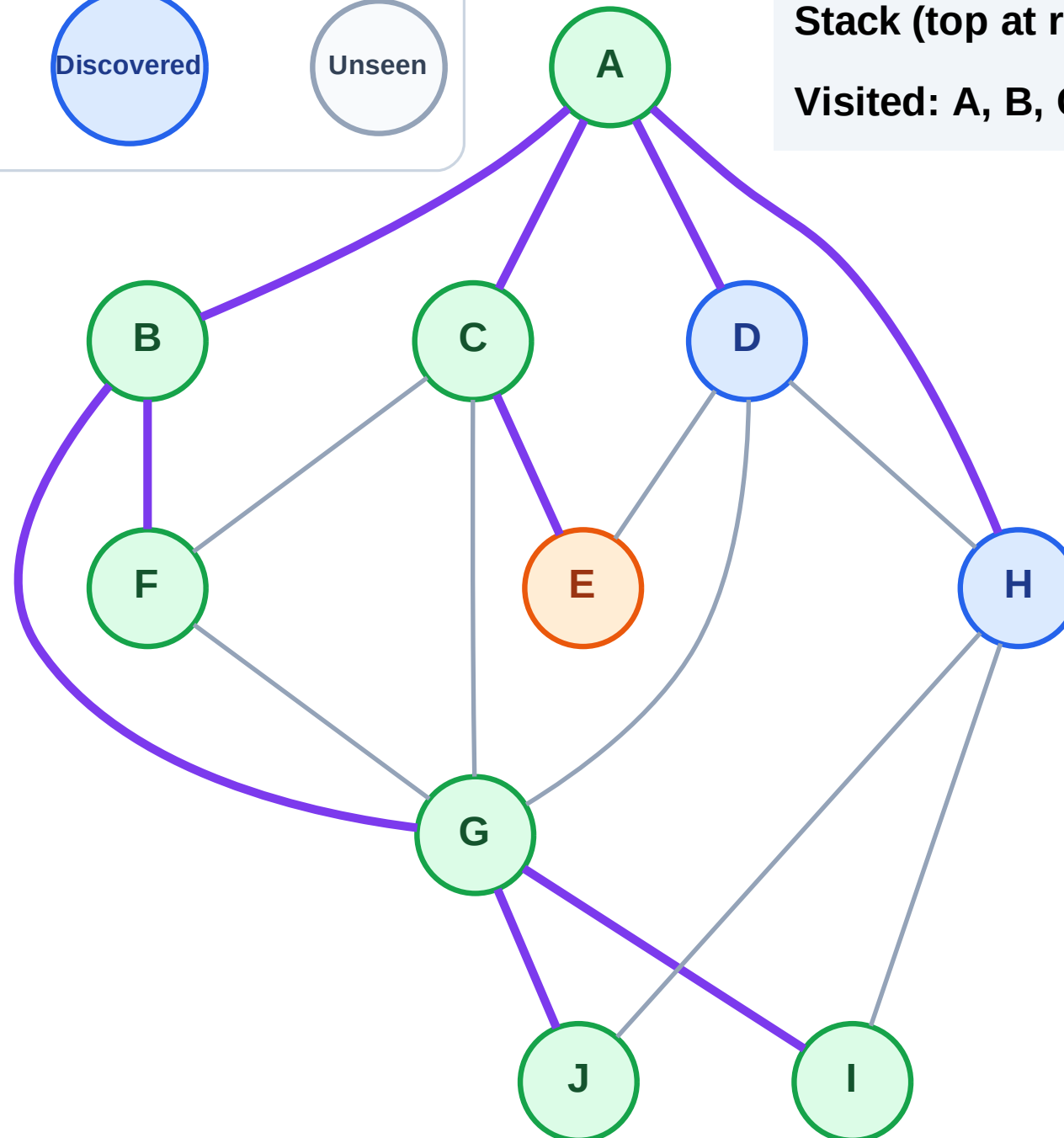
Processing

Discovered

Unseen

Stack (top at right): H | D

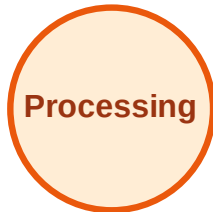
Visited: A, B, C, F, G, I, J



DFS Traversal

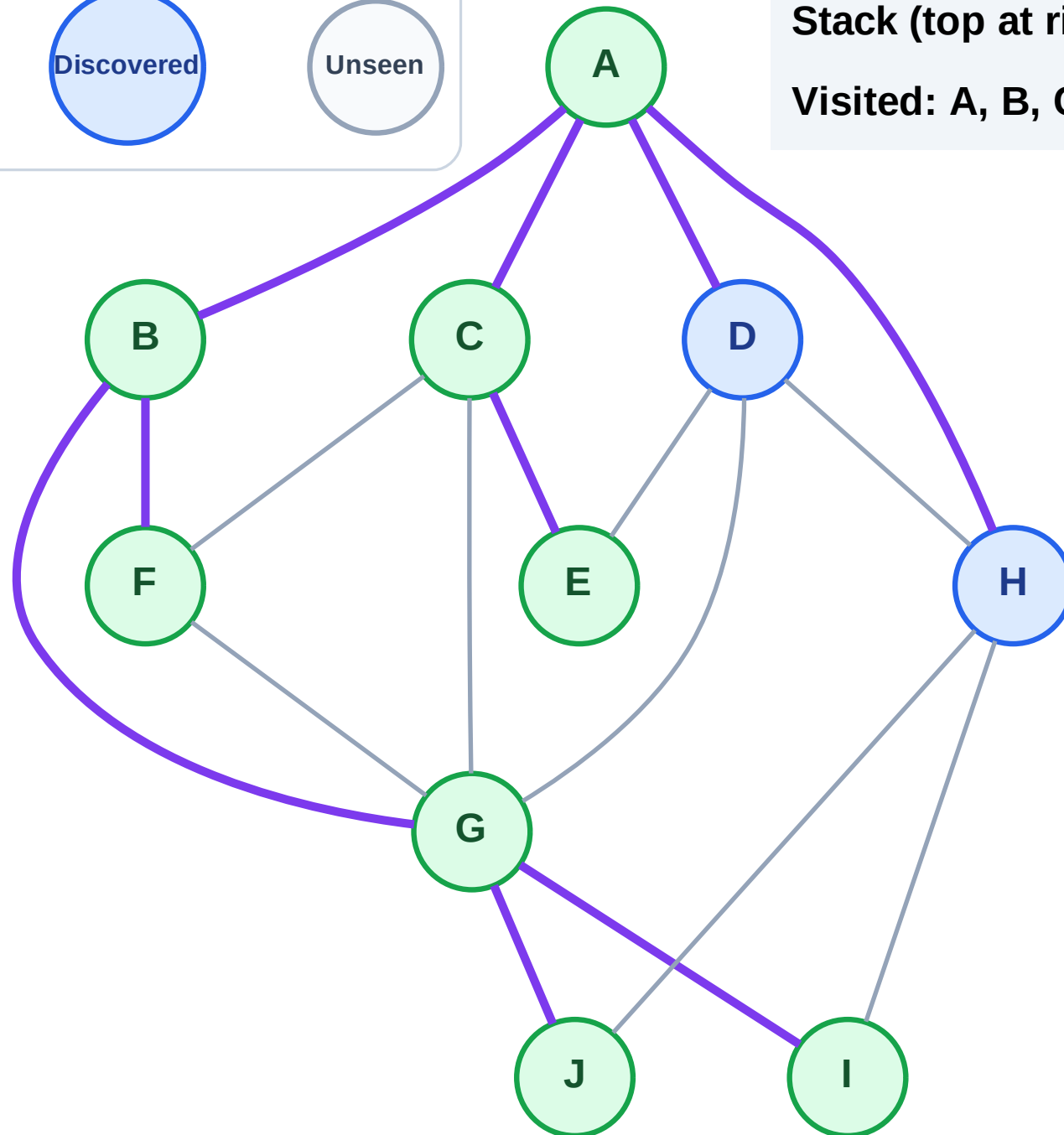
Step 17: No new neighbors from E

Legend



Stack (top at right): H | D

Visited: A, B, C, E, F, G, I, J



DFS Traversal

Step 18: Process node D

Legend

Complete

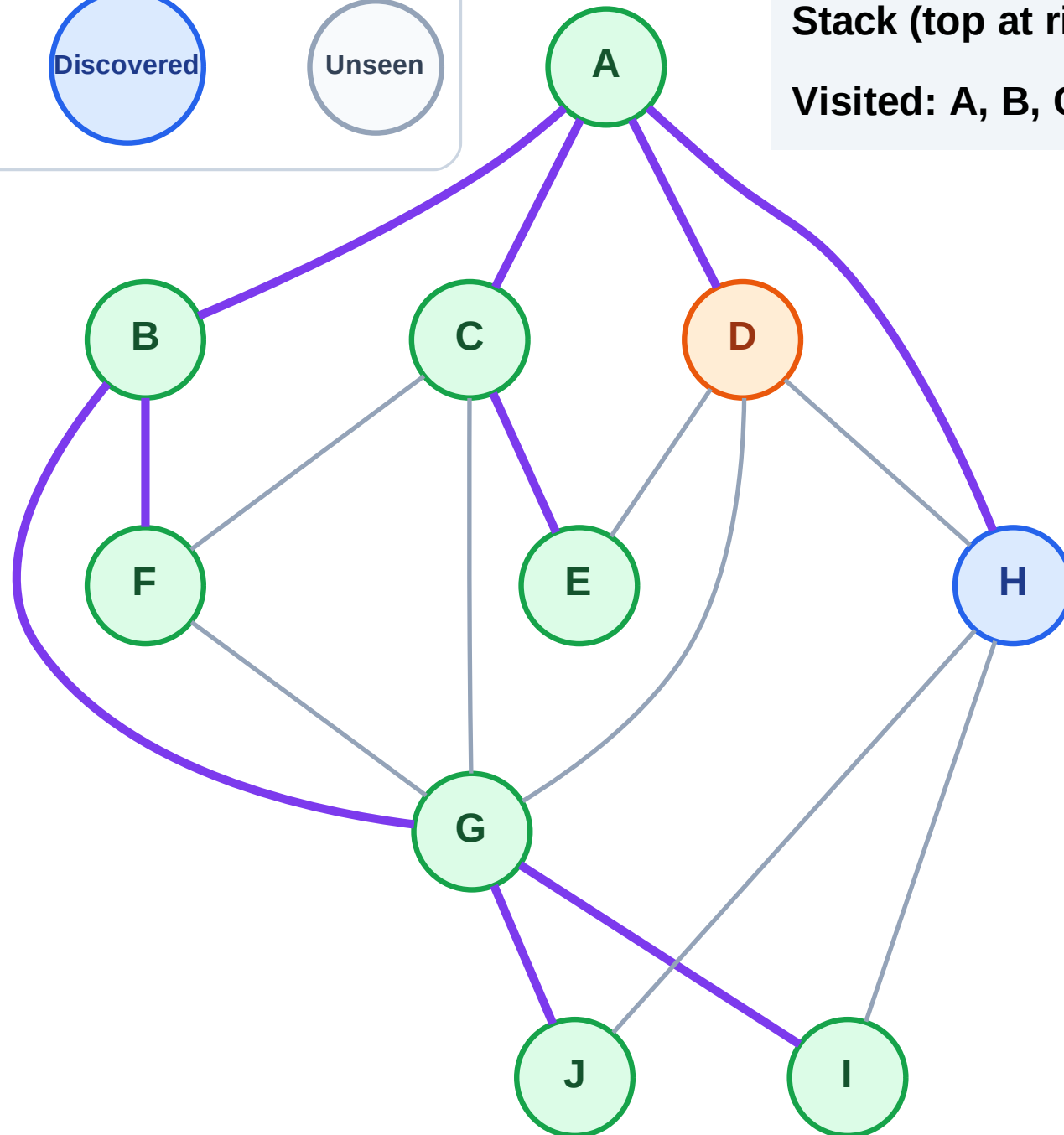
Processing

Discovered

Unseen

Stack (top at right): H

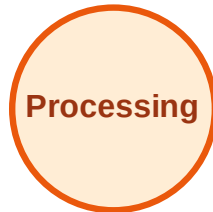
Visited: A, B, C, E, F, G, I, J



DFS Traversal

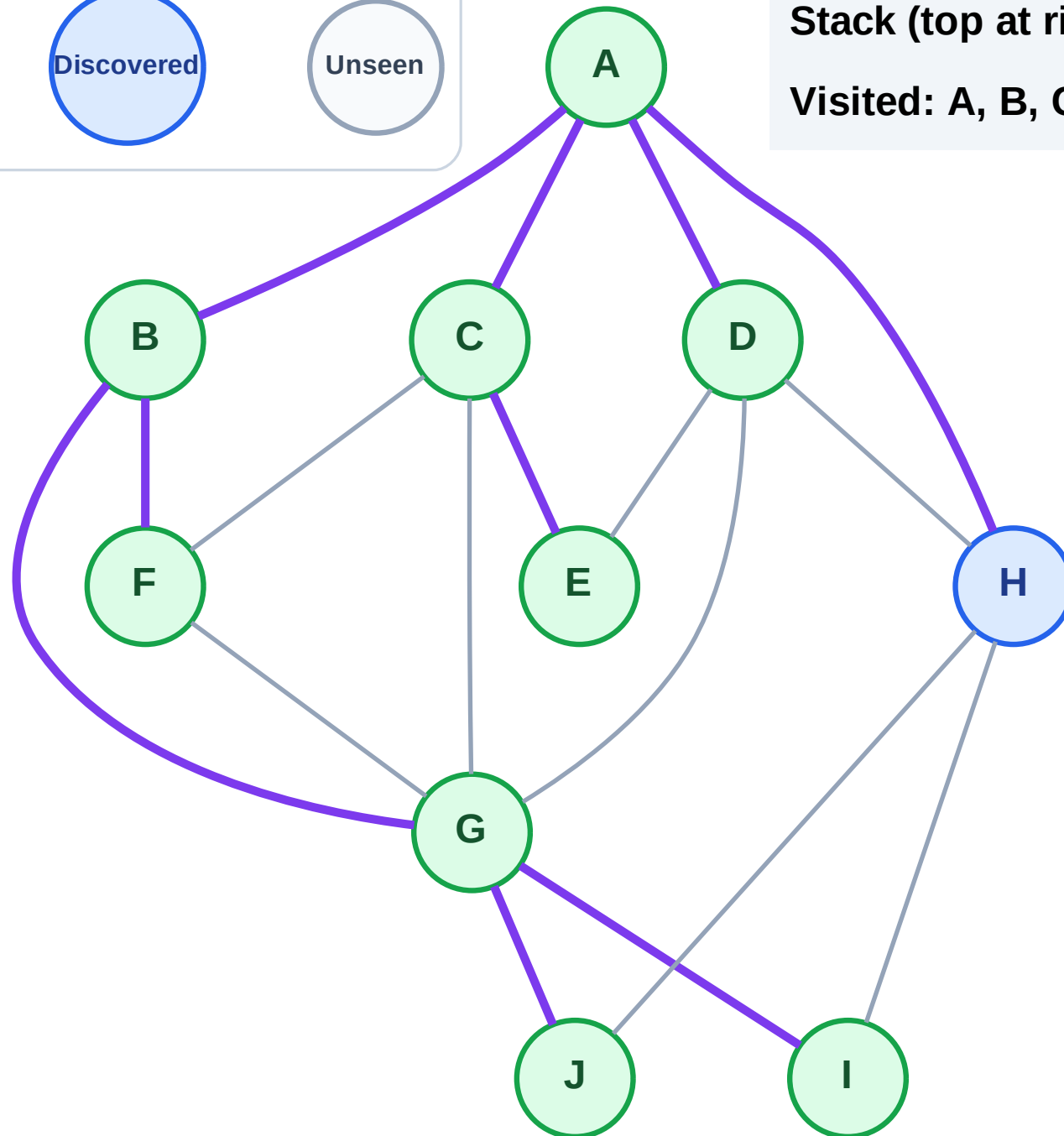
Step 19: No new neighbors from D

Legend



Stack (top at right): H

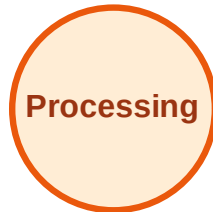
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

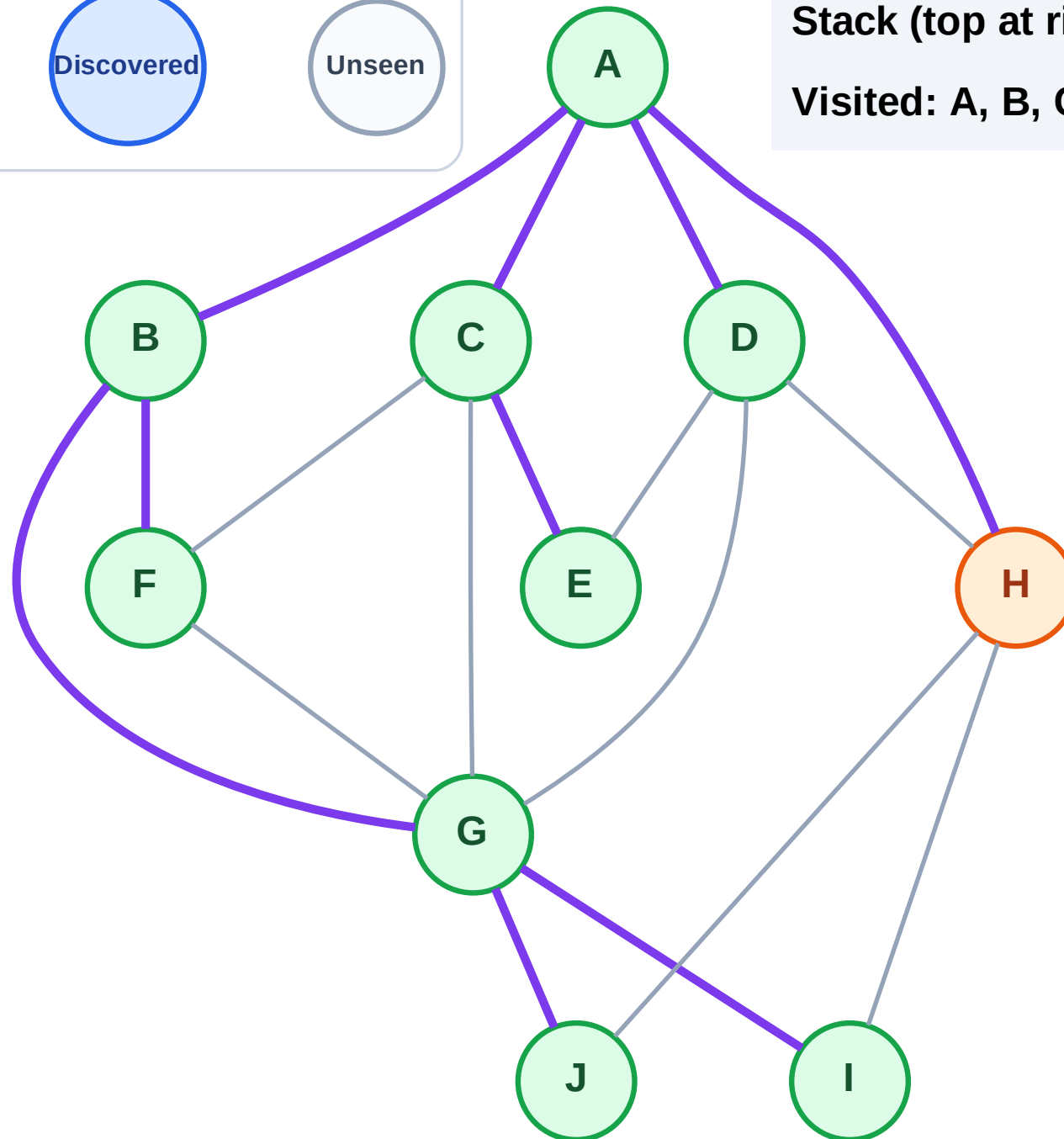
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

